CALCULUS

Volume I

One-Variable Calculus, with an Introduction to Linear Algebra

Second Edition

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Part I

Introduction

1
INTRODUCTION

1.1 The Two Basic Concepts of Calculus

Calculus consists of two main branches: **integral calculus** and **differential calculus**.

1. **The area problem (Integral Calculus):** To assign a number which measures the area of a region bounded by a curve.

2. **The tangent problem (Differential Calculus):** To assign a number which measures the steepness of a tangent line to a curve.

2

SET THEORY

2.1 Set Theory

DEFINITION 2.1. Set A *set* is a collection of objects, called *elements* or *members* of the set.
DEFINITION 2.2. Subset A set A is called a *subset* of a set B, denoted by $A \subseteq B$, if every element of A also belongs to B.
DEFINITION 2.3. Union The *union* of two sets A and B, denoted by $A \cup B$, is the set of all elements that belong to A or to B (or to both).
DEFINITION 2.4. Intersection The *intersection* of two sets A and B, denoted by $A \cap B$, is the set of all elements that belong to both A and B.
DEFINITION 2.5. Complement The *complement* of a set A relative to a set B, denoted by $B - A$ or $B \setminus A$, is the set of all elements that belong to B but not to A.

3

AXIOMS FOR REAL NUMBERS

3.1 A Set of Axioms for the Real-Number System

3.1.1 The Field Axioms

Axiom 3.1. Commutative Laws. $x + y = y + x, \quad xy = yx.$
Axiom 3.2. Associative Laws. $x + (y + z) = (x + y) + z, \quad x(yz) = (xy)z.$
Axiom 3.3. Distributive Law. $x(y + z) = xy + xz.$
Axiom 3.4. Existence of Identity Elements. There exist two distinct real numbers, which we denote by 0 and 1, such that for every real x we have $x + 0 = x$ and $1 \cdot x = x$.
Axiom 3.5. Existence of Negatives. For every real number x there is a real number y such that $x + y = 0$.
Axiom 3.6. Existence of Reciprocals. For every real number $x \neq 0$ there is a real number y such that $xy = 1$.
Note: The numbers 0 and 1 in Axioms 5 and 6 are those of Axiom 4.
THEOREM 3.1.

(a) If $a + b = a + c$, then $b = c$. *(Cancellation law for addition)*

(b) If $ab = ac$ and $a \neq 0$, then $b = c$. *(Cancellation law for multiplication)*

Proof. (a) Given $a + b = a + c$. By Axiom 5 there exists a number y such that $a + y = 0$. Adding y to both sides, we get $y + (a + b) = y + (a + c)$. By associativity, $(y + a) + b = (y + a) + c$, so $0 + b = 0 + c$, thus $b = c$.

(b) Given $ab = ac$ and $a \neq 0$. By Axiom 6 there exists a number z such that $az = 1$. Multiplying both sides by z, we get $z(ab) = z(ac)$. By associativity, $(za)b = (za)c$, so $1 \cdot b = 1 \cdot c$, thus $b = c$. □

THEOREM 3.2. For any real numbers a and b, there is exactly one real number x such that $a + x = b$.

Proof. By Axiom 5, there exists y such that $a + y = 0$. Let $x = y + b$. Then $a + x = a + (y + b) = (a + y) + b = 0 + b = b$. For uniqueness, suppose $a + x = b$ and $a + x' = b$. Then $a + x = a + x'$, so by cancellation, $x = x'$. □

THEOREM 3.3. For any real numbers a and b with $a \neq 0$, there is exactly one real number x such that $ax = b$.

Proof. Similar to Theorem 1.2, using Axiom 6. □

THEOREM 3.4. If $a \cdot 0 = 0$ for every real number a.

Proof. We have $a \cdot 0 + a \cdot 0 = a(0 + 0) = a \cdot 0 = a \cdot 0 + 0$. By cancellation, $a \cdot 0 = 0$. □

THEOREM 3.5. If $ab = 0$, then $a = 0$ or $b = 0$.

Proof. Suppose $ab = 0$ and $a \neq 0$. Then $b = 1 \cdot b = (a^{-1}a)b = a^{-1}(ab) = a^{-1} \cdot 0 = 0$. □

3.1.2 The Order Axioms

Let $\mathbb{R}^+$ denote the set of positive real numbers.
Axiom 3.7. Closure of $\mathbb{R}^+$. If x and y are in $\mathbb{R}^+$, so are $x+y$ and xy.
Axiom 3.8. Trichotomy. For every real $x \neq 0$, either $x \in \mathbb{R}^+$ or $-x \in \mathbb{R}^+$, but not both.
Axiom 3.9. Nonmembership of Zero. $0 \notin \mathbb{R}^+$.
DEFINITION 3.1. Inequality $a < b$ means $b - a \in \mathbb{R}^+$.
THEOREM 3.6.

(a) If $a < b$ and $b < c$, then $a < c$. *(Transitivity)*

(b) If $a < b$, then $a + c < b + c$.

(c) If $a < b$ and $c > 0$, then $ac < bc$.

Proof. (a) If $b-a \in \mathbb{R}^+$ and $c-b \in \mathbb{R}^+$, then $(b-a)+(c-b) = c-a \in \mathbb{R}^+$ by closure.
(b) $(b+c) - (a+c) = b - a \in \mathbb{R}^+$.
(c) If $c > 0$, then $c \in \mathbb{R}^+$, so $bc - ac = (b-a)c \in \mathbb{R}^+$ by closure. □

3.1.3 Integers and Rational Numbers

DEFINITION 3.2. Inductive Set A set of real numbers is called an *inductive set* if (a) the number 1 is in the set, and (b) for every x in the set, the number $x+1$ is also in the set.
DEFINITION 3.3. Positive Integers ($\mathbb{P}$) A real number is called a *positive integer* if it belongs to every inductive set. $\mathbb{P}$ is the smallest inductive set.
DEFINITION 3.4. Rational Numbers ($\mathbb{Q}$) Quotients of integers a/b (where $b \neq 0$). The set $\mathbb{Q}$ contains $\mathbb{Z}$ as a subset and forms an ordered field.

3.1.4 Upper Bounds and Supremum

DEFINITION 3.5. Upper Bound Let S be a nonempty set of real numbers. If there is a number B such that $x \leq B$ for every x in S, then S is said to be *bounded above* by B. The number B is called an *upper bound* for S.
DEFINITION 3.6. Maximum Element If an upper bound B is also a member of S, then B is called the *largest member* or the *maximum element* of S, denoted by $B = \max S$.
DEFINITION 3.7. Least Upper Bound (Supremum) A number B is called a *least upper bound* of a nonempty set S if: (a) B is an upper bound for S; (b) No number less than B is an upper bound for S. We write $B = \sup S$.
Axiom 3.10. Least-Upper-Bound Axiom (Completeness Axiom). Every nonempty set S of real numbers which is bounded above has a supremum; that is, there is a real number B such that $B = \sup S$.
DEFINITION 3.8. Lower Bound A set S is bounded below if there exists a number L such that $L \leq x$ for all $x \in S$.
DEFINITION 3.9. Minimum Element If a lower bound L is also a member of S, then L is called the *smallest member* or *minimum element* of S, denoted by $L = \min S$.

DEFINITION 3.10. Greatest Lower Bound (Infimum) A number L is called a *greatest lower bound* (or *infimum*) of S if: (a) L is a lower bound for S; (b) no number greater than L is a lower bound for S. We denote it by $\inf S$.

THEOREM 3.7. Every nonempty set S that is bounded below has a greatest lower bound; that is, there is a real number L such that $L = \inf S$.

Proof. Let $-S$ denote the set of negatives of numbers in S. Then $-S$ is nonempty and bounded above. By the completeness axiom, there is a number B which is a supremum for $-S$. Then $-B = \inf S$. □

3.1.5 The Archimedean Property

THEOREM 3.8. The set $\mathbb{P}$ of positive integers $1, 2, 3, \ldots$ is unbounded above.

Proof. Assume $\mathbb{P}$ is bounded above. Since $\mathbb{P}$ is nonempty, by the completeness axiom $\mathbb{P}$ has a least upper bound, say b. The number $b - 1$, being less than b, cannot be an upper bound for $\mathbb{P}$. Hence, there is at least one positive integer n such that $n > b - 1$. For this n we have $n + 1 > b$. Since $n + 1$ is in $\mathbb{P}$, this contradicts the fact that b is an upper bound for $\mathbb{P}$. □

THEOREM 3.9. Archimedean Property If $x > 0$ and if y is an arbitrary real number, there exists a positive integer n such that $nx > y$.

Proof. If $nx \leq y$ for all n, then $n \leq y/x$ for all n, contradicting the unboundedness of $\mathbb{P}$. □

THEOREM 3.10. If three real numbers a, x, and y satisfy the inequalities

$$a \leq x \leq a + \frac{y}{n}$$

for every integer $n \geq 1$, then $x = a$.

Proof. If $x > a$, by the Archimedean property there is a positive integer n satisfying $n(x - a) > y$, contradicting $x \leq a + y/n$. □

4

INDUCTION AND SUMMATION

4.1 Mathematical Induction, Summation Notation, and Related Topics

THEOREM 4.1. Principle of Mathematical Induction Let $P(n)$ be a proposition about the positive integer n. If:

(a) $P(1)$ is true, and

(b) Whenever $P(k)$ is true for some positive integer k, $P(k+1)$ is also true,

then $P(n)$ is true for all positive integers n.

Proof. Let S be the set of positive integers for which $P(n)$ is true. By (a), $1 \in S$. By (b), if $k \in S$, then $k+1 \in S$. Thus S is an inductive set, so $\mathbb{P} \subseteq S$. Since $S \subseteq \mathbb{P}$ by definition, $S = \mathbb{P}$. □

DEFINITION 4.1. Summation Notation If $a_1, a_2, \ldots, a_n$ are real numbers, we define:

$$\sum_{k=1}^{n} a_k = a_1 + a_2 + \cdots + a_n$$

THEOREM 4.2. Properties of Summation

(a) $\displaystyle\sum_{k=1}^{n}(a_k + b_k) = \sum_{k=1}^{n} a_k + \sum_{k=1}^{n} b_k$

(b) $\displaystyle\sum_{k=1}^{n} ca_k = c\sum_{k=1}^{n} a_k$ for any constant c

(c) $\displaystyle\sum_{k=1}^{n} c = nc$ for any constant c

Proof. Follows directly from the commutative, associative, and distributive laws. □

DEFINITION 4.2. Absolute Value For any real number x, the *absolute value* of x, denoted by $|x|$, is defined by:

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

THEOREM 4.3. Properties of Absolute Value

(a) $|x| \geq 0$ for all x, and $|x| = 0$ if and only if $x = 0$

(b) $|-x| = |x|$ for all x

(c) $|xy| = |x||y|$ for all x, y

(d) $|x/y| = |x|/|y|$ for all x and $y \neq 0$

Proof. Immediate from the definition. □

THEOREM 4.4. Triangle Inequality For all real numbers x and y:

$$|x + y| \leq |x| + |y|$$

Proof. We have $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$. Adding these inequalities gives $-(|x| + |y|) \leq x + y \leq |x| + |y|$. This is equivalent to $|x + y| \leq |x| + |y|$. □

THEOREM 4.5. For all real numbers x and y:

$$||x| - |y|| \leq |x - y|$$

Proof. We have $|x| = |(x - y) + y| \leq |x - y| + |y|$, so $|x| - |y| \leq |x - y|$. Interchanging x and y gives $|y| - |x| \leq |y - x| = |x - y|$. Combining these gives $||x| - |y|| \leq |x - y|$. □

THEOREM 4.6. Bernoulli's Inequality If $x > -1$ and n is a positive integer, then

$$(1 + x)^n \geq 1 + nx$$

with equality if and only if $n = 1$ or $x = 0$.

Proof. By induction. For $n = 1$, equality holds. Assume $(1 + x)^k \geq 1 + kx$. Then $(1 + x)^{k+1} = (1 + x)^k(1 + x) \geq (1 + kx)(1 + x) = 1 + (k + 1)x + kx^2 \geq 1 + (k + 1)x$. Equality holds only if $k = 0$ or $x = 0$. □

THEOREM 4.7. Cauchy-Schwarz Inequality For any real numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$:

$$\left(\sum_{k=1}^{n} a_k b_k\right)^2 \leq \left(\sum_{k=1}^{n} a_k^2\right)\left(\sum_{k=1}^{n} b_k^2\right)$$

Proof. Consider $\sum_{k=1}^{n}(a_k x + b_k)^2 = x^2 \sum a_k^2 + 2x \sum a_k b_k + \sum b_k^2 \geq 0$ for all x. This quadratic in x is nonnegative, so its discriminant is nonpositive: $4(\sum a_k b_k)^2 - 4(\sum a_k^2)(\sum b_k^2) \leq 0$. □

Part II

Integral Calculus

5

THE INTEGRAL

5.1 Functions

DEFINITION 5.1. Function A *function* f from a set A to a set B is a set of ordered pairs (x, y) with $x \in A$ and $y \in B$ such that no two distinct pairs have the same first element. If $(x, y) \in f$, we write $y = f(x)$. The set A is called the *domain* of f, and the set of all y such that $y = f(x)$ for some x is called the *range* of f.

5.2 The Concept of Area as a Set Function

DEFINITION 5.2. Area Function An *area function* is a nonnegative set function a defined on a collection of sets in the plane, satisfying the following properties:

(1) *Additive property:* If S and T are in the collection and $S \cap T = \emptyset$, then $a(S \cup T) = a(S) + a(T)$.

(2) *Comparison property:* If S and T are in the collection with $S \subseteq T$, then $a(S) \leq a(T)$.

5.3 Intervals and Ordinate Sets

DEFINITION 5.3. Interval Let $a < b$. The set of all x satisfying $a < x < b$ is called the *open interval* (a, b). The set of all x satisfying $a \leq x \leq b$ is called the *closed interval* $[a, b]$. The sets $[a, b)$ and $(a, b]$ are called *half-open intervals.*

DEFINITION 5.4. Ordinate Set Let f be a nonnegative function defined on $[a, b]$. The *ordinate set* of f over $[a, b]$ is the set:

$$S = \{(x, y) \mid a \leq x \leq b,\ 0 \leq y \leq f(x)\}$$

5.4 Partitions and Step Functions

DEFINITION 5.5. Partition A *partition* P of an interval $[a, b]$ is a set of points $P = \{x_0, x_1, \ldots, x_n\}$ such that $a = x_0 < x_1 < \cdots < x_n = b$. The k-th subinterval is $[x_{k-1}, x_k]$ and has length $\Delta x_k = x_k - x_{k-1}$.

DEFINITION 5.6. Step Function A function s defined on $[a, b]$ is called a *step function* if there exists a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ such that s is constant on each open subinterval (x_{k-1}, x_k).

5.5 The Integral of Step Functions

DEFINITION 5.7. Integral of a Step Function Let s be a step function defined on $[a, b]$ which takes the constant value s_k on the k-th open subinterval (x_{k-1}, x_k) of a partition P. The *integral* of s from a to b is defined by:

$$\int_a^b s(x)\,dx = \sum_{k=1}^{n} s_k \Delta x_k$$

THEOREM 5.1. The integral of a step function is independent of the choice of partition.

Proof. Suppose s is constant on the open subintervals of partition P and also on the open subintervals of a refinement P'. The values s_k assigned to each subinterval are the same whether we use P or P', because s is constant on any subinterval where it is defined. The sum $\sum s_k \Delta x_k$ gives the same value for both partitions. □

THEOREM 5.2. Additivity with Respect to the Interval If s is a step function on $[a, b]$ and $a < c < b$, then:

$$\int_a^b s(x)\,dx = \int_a^c s(x)\,dx + \int_c^b s(x)\,dx$$

Proof. We can choose a partition that includes c as a partition point. Then the sum $\sum_{k=1}^{n} s_k \Delta x_k$ splits naturally into two parts: those subintervals in $[a, c]$ and those in $[c, b]$. □

THEOREM 5.3. Homogeneous Property For any step function s and any constant c:

$$\int_a^b c \cdot s(x)\,dx = c \int_a^b s(x)\,dx$$

Proof. If $s(x) = s_k$ on (x_{k-1}, x_k), then $c \cdot s(x) = c \cdot s_k$ on the same interval. Therefore:

$$\int_a^b c \cdot s(x)\,dx = \sum_{k=1}^{n} c \cdot s_k \Delta x_k = c \sum_{k=1}^{n} s_k \Delta x_k = c \int_a^b s(x)\,dx$$

□

THEOREM 5.4. Additive Property If s and t are step functions on $[a, b]$, then:

$$\int_a^b [s(x) + t(x)]\,dx = \int_a^b s(x)\,dx + \int_a^b t(x)\,dx$$

Proof. Let P be a partition fine enough so that both s and t are constant on each open subinterval. If $s(x) = s_k$ and $t(x) = t_k$ on (x_{k-1}, x_k), then $s(x) + t(x) = s_k + t_k$ on this interval. Therefore:

$$\int_a^b [s(x) + t(x)]\,dx = \sum_{k=1}^{n} (s_k + t_k)\Delta x_k = \sum_{k=1}^{n} s_k \Delta x_k + \sum_{k=1}^{n} t_k \Delta x_k = \int_a^b s(x)\,dx + \int_a^b t(x)\,dx$$

□

THEOREM 5.5. Comparison Theorem for Step Functions If s and t are step functions on $[a, b]$ with $s(x) \leq t(x)$ for all $x \in [a, b]$, then:

$$\int_a^b s(x)\,dx \leq \int_a^b t(x)\,dx$$

Proof. Since $s(x) \leq t(x)$ on each subinterval, we have $s_k \leq t_k$ for each k. Therefore $s_k \Delta x_k \leq t_k \Delta x_k$, and summing gives the result. □

5.6 Upper and Lower Integrals

DEFINITION 5.8. Upper and Lower Step Functions Let f be a bounded function on $[a, b]$. A step function s is said to be *below* f (or a *lower step function*) if $s(x) \leq f(x)$ for all $x \in [a, b]$. A step function t is said to be *above* f (or an *upper step function*) if $f(x) \leq t(x)$ for all $x \in [a, b]$.

DEFINITION 5.9. Lower and Upper Integrals Let f be a bounded function on $[a, b]$. The *lower integral* of f is defined by:

$$I(f) = \sup \left\{ \int_a^b s(x)\, dx \mid s \text{ is a step function below } f \right\}$$

The *upper integral* of f is defined by:

$$\bar{I}(f) = \inf \left\{ \int_a^b t(x)\, dx \mid t \text{ is a step function above } f \right\}$$

THEOREM 5.6. For any bounded function f on $[a, b]$, we have $I(f) \leq \bar{I}(f)$.

Proof. If s is below f and t is above f, then $s(x) \leq f(x) \leq t(x)$ for all x, so $s(x) \leq t(x)$. By the comparison theorem for step functions, $\int_a^b s \leq \int_a^b t$. Taking supremum over all s below f gives $I(f) \leq \int_a^b t$ for any t above f. Taking infimum over all t above f gives $I(f) \leq \bar{I}(f)$. □

5.7 The Definition of the Integral for General Functions

DEFINITION 5.10. Integrable Function A bounded function f defined on $[a, b]$ is said to be *integrable* on $[a, b]$ if $I(f) = \bar{I}(f)$. In this case, the common value is called the *integral* of f from a to b, denoted by $\int_a^b f(x)\, dx$.

DEFINITION 5.11. Area of an Ordinate Set If f is nonnegative and integrable on $[a, b]$, the *area* of the ordinate set of f over $[a, b]$ is defined to be $\int_a^b f(x)\, dx$.

5.8 Monotonic and Piecewise Monotonic Functions

DEFINITION 5.12. Monotonic Function A function f defined on an interval is said to be *increasing* on the interval if $f(x) \leq f(y)$ whenever $x < y$. It is said to be *decreasing* if $f(x) \geq f(y)$ whenever $x < y$. A function is *monotonic* if it is either increasing or decreasing.

DEFINITION 5.13. Piecewise Monotonic Function A function f defined on $[a, b]$ is called *piecewise monotonic* if there exists a partition $P = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$ such that f is monotonic on each open subinterval (x_{k-1}, x_k).

THEOREM 5.7. Integrability of Bounded Monotonic Functions If f is bounded and monotonic on $[a, b]$, then f is integrable on $[a, b]$.

Proof. Assume f is increasing (the proof for decreasing is similar). Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$ into n equal subintervals, each of length $\Delta x = (b-a)/n$. Define step functions s and t as follows: On the k-th open subinterval (x_{k-1}, x_k), let $s(x) = f(x_{k-1})$ and $t(x) = f(x_k)$. Since f is increasing, $s(x) \leq f(x) \leq t(x)$ on each subinterval.

The integrals of these step functions are:

$$\int_a^b s = \sum_{k=1}^{n} f(x_{k-1})\Delta x \quad \text{and} \quad \int_a^b t = \sum_{k=1}^{n} f(x_k)\Delta x$$

Their difference is:

$$\int_a^b t - \int_a^b s = \sum_{k=1}^{n}[f(x_k) - f(x_{k-1})]\Delta x = \Delta x \sum_{k=1}^{n}[f(x_k) - f(x_{k-1})]$$

This is a telescoping sum equal to $\Delta x[f(b) - f(a)] = \frac{b-a}{n}[f(b) - f(a)]$.

Given $\epsilon > 0$, choose n large enough so that $\frac{b-a}{n}[f(b) - f(a)] < \epsilon$. Then $\int_a^b t - \int_a^b s < \epsilon$. Since $I(f)$ and $\bar{I}(f)$ lie between these two values, we have $\bar{I}(f) - I(f) < \epsilon$. Since ϵ is arbitrary, $I(f) = \bar{I}(f)$, so f is integrable. □

5.9 Calculation of the Integral

THEOREM 5.8. If f is constant on $[a, b]$ with $f(x) = c$ for all x, then:

$$\int_a^b f(x)\,dx = c(b - a)$$

Proof. The function f itself is a step function with respect to any partition, so $\int_a^b f = c(b - a)$. □

THEOREM 5.9. If $f(x) = x$ for all $x \in [a, b]$, then:

$$\int_a^b x\,dx = \frac{b^2 - a^2}{2}$$

Proof. Consider the partition $P = \{x_0, x_1, \ldots, x_n\}$ where $x_k = a + \frac{k(b-a)}{n}$. Then $\Delta x = \frac{b-a}{n}$ and $x_k = a + k\Delta x$.

For the lower step function s with $s(x) = x_{k-1}$ on (x_{k-1}, x_k):

$$\int_a^b s = \sum_{k=1}^{n} x_{k-1}\Delta x = \sum_{k=1}^{n}[a + (k-1)\Delta x]\Delta x = a(b - a) + (\Delta x)^2 \sum_{k=1}^{n}(k - 1)$$

Since $\sum_{k=1}^{n}(k - 1) = \frac{n(n-1)}{2}$, we get:

$$\int_a^b s = a(b - a) + \frac{(b - a)^2}{n^2} \cdot \frac{n(n - 1)}{2} = a(b - a) + \frac{(b - a)^2}{2}\left(1 - \frac{1}{n}\right)$$

Taking $n \to \infty$: $I(f) = a(b - a) + \frac{(b-a)^2}{2} = \frac{b^2-a^2}{2}$.

Similarly, for the upper step function, $\bar{I}(f) = \frac{b^2-a^2}{2}$. Thus $\int_a^b x\,dx = \frac{b^2-a^2}{2}$. □

THEOREM 5.10. If p is a positive integer and $b > 0$, then:

$$\int_0^b x^p\,dx = \frac{b^{p+1}}{p + 1}$$

Proof. Use the partition $x_k = \frac{kb}{n}$ for $k = 0, 1, \ldots, n$, so $\Delta x = \frac{b}{n}$. For the lower step function with $s(x) = x_{k-1}^p$ on (x_{k-1}, x_k):

$$\int_0^b s = \sum_{k=1}^{n}\left(\frac{(k - 1)b}{n}\right)^p \cdot \frac{b}{n} = \frac{b^{p+1}}{n^{p+1}}\sum_{k=1}^{n}(k - 1)^p = \frac{b^{p+1}}{n^{p+1}}\sum_{j=0}^{n-1} j^p$$

Using the formula $\sum_{j=1}^{n-1} j^p = \frac{(n-1)^{p+1}}{p+1} + O(n^p)$, as $n \to \infty$:

$$\int_0^b s \to \frac{b^{p+1}}{p + 1}$$

Similarly for the upper integral, giving $\int_0^b x^p\,dx = \frac{b^{p+1}}{p+1}$. □

5.10 Basic Properties of the Integral

THEOREM 5.11. Additivity Property If f is integrable on $[a, b]$ and $a < c < b$, then f is integrable on $[a, c]$ and $[c, b]$, and:

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

Proof. Let s_1 be a step function below f on $[a, c]$ and s_2 a step function below f on $[c, b]$. Then s defined by $s(x) = s_1(x)$ on $[a, c)$ and $s(x) = s_2(x)$ on $[c, b]$ is a step function below f on $[a, b]$, and:

$$\int_a^b s = \int_a^c s_1 + \int_c^b s_2$$

Taking supremum over all such s_1 and s_2 gives $I_{[a,b]}(f) \geq I_{[a,c]}(f) + I_{[c,b]}(f)$. Similarly, $\bar{I}_{[a,b]}(f) \leq \bar{I}_{[a,c]}(f) + \bar{I}_{[c,b]}(f)$. Since f is integrable on $[a, b]$, these give equality. □

THEOREM 5.12. Homogeneous Property If f is integrable on $[a, b]$ and c is any real number, then cf is integrable on $[a, b]$ and:

$$\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$$

Proof. For $c > 0$: If $s \leq f \leq t$, then $cs \leq cf \leq ct$. Also $\int_a^b cs = c\int_a^b s$ and $\int_a^b ct = c\int_a^b t$. Taking sup and inf gives the result. For $c < 0$, the inequalities reverse but the conclusion still holds. □

THEOREM 5.13. Linearity Property If f and g are integrable on $[a, b]$, then $f+g$ is integrable on $[a, b]$ and:

$$\int_a^b [f(x) + g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

Proof. Given $\epsilon > 0$, choose step functions s_1, t_1 for f and s_2, t_2 for g such that:

$$\int_a^b s_1 > I(f) - \frac{\epsilon}{2}, \quad \int_a^b t_1 < \bar{I}(f) + \frac{\epsilon}{2}$$

$$\int_a^b s_2 > I(g) - \frac{\epsilon}{2}, \quad \int_a^b t_2 < \bar{I}(g) + \frac{\epsilon}{2}$$

Then $s_1 + s_2 \leq f + g \leq t_1 + t_2$, and:

$$\int_a^b (t_1 + t_2) - \int_a^b (s_1 + s_2) = \left(\int_a^b t_1 - \int_a^b s_1\right) + \left(\int_a^b t_2 - \int_a^b s_2\right) < \epsilon$$

This shows $f+g$ is integrable. Moreover, taking limits gives the linearity formula. □

THEOREM 5.14. Comparison Theorem If f and g are integrable on $[a, b]$ with $f(x) \leq g(x)$ for all $x \in [a, b]$, then:

$$\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx$$

Proof. If s is a step function below f, then s is also below g. Therefore:

$$\int_a^b s \leq I(g) = \int_a^b g$$

Taking supremum over all s below f gives $\int_a^b f \leq \int_a^b g$. $\square$

THEOREM 5.15. If f is integrable on $[a, b]$ and $m \leq f(x) \leq M$ for all $x \in [a, b]$, then:

$$m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a)$$

Proof. Apply the comparison theorem with the constant functions m and M. $\square$

6

APPLICATIONS OF INTEGRATION

6.1 Area Between Two Curves

DEFINITION 6.1. Area Between Two Curves If f and g are integrable on $[a, b]$ with $f(x) \leq g(x)$ for all $x \in [a, b]$, the area A of the region between the graphs of f and g over $[a, b]$ is:

$$A = \int_a^b [g(x) - f(x)]\, dx$$

6.2 The Trigonometric Functions

THEOREM 6.1. Integration Formulas for Sine and Cosine For any real numbers a and b:

$$\int_a^b \cos x\, dx = \sin b - \sin a$$

$$\int_a^b \sin x\, dx = -(\cos b - \cos a) = \cos a - \cos b$$

Proof. These formulas are derived from the fundamental properties of the sine and cosine functions and their derivatives. The sine and cosine functions are defined geometrically using the unit circle, and their integral formulas follow from the fact that $\frac{d}{dx} \sin x = \cos x$ and $\frac{d}{dx} \cos x = -\sin x$. □

6.3 Polar Coordinates

DEFINITION 6.2. Polar Coordinates The *polar coordinates* of a point P in the plane are an ordered pair of numbers (r, θ), where r is the distance from P to the origin (pole), and θ is the angle that the ray OP makes with the positive x-axis (polar axis).

THEOREM 6.2. Area in Polar Coordinates If a function f is nonnegative and integrable on $[a, b]$ where $0 \leq b - a \leq 2\pi$, the area A of the radial set of f over $[a, b]$ is:

$$A = \frac{1}{2} \int_a^b [f(\theta)]^2\, d\theta$$

Proof. The area of a circular sector of radius r and angle $\Delta\theta$ is $\frac{1}{2} r^2 \Delta\theta$. For a general function $r = f(\theta)$, we partition the interval $[a, b]$ and approximate the area by summing the areas of circular sectors. Taking the limit as the partition becomes finer gives the integral formula. □

6.4 Volume by Integration

DEFINITION 6.3. Volume by Cross-Sectional Area Let S be a solid. A cross-section of S perpendicular to a given axis is the intersection of S with a plane perpendicular to that axis. If the cross-sectional area at position x is $A(x)$, and A is integrable on $[a, b]$, the volume V of S is:

$$V = \int_a^b A(x)\,dx$$

THEOREM 6.3. Volume of Solid of Revolution If f is integrable on $[a, b]$ and $f(x) \geq 0$ for all $x \in [a, b]$, the volume V of the solid obtained by rotating the ordinate set of f about the x-axis is:

$$V = \pi \int_a^b [f(x)]^2\,dx$$

Proof. The cross-section at position x is a circle of radius $f(x)$, so $A(x) = \pi[f(x)]^2$. By the definition of volume by cross-sectional area, $V = \int_a^b A(x)\,dx = \pi \int_a^b [f(x)]^2\,dx$. □

6.5 Work

DEFINITION 6.4. Work If a force $f(x)$ moves an object from $x = a$ to $x = b$, the *work* W done by the force is:

$$W = \int_a^b f(x)\,dx$$

provided f is integrable on $[a, b]$.

6.6 Average Value of a Function

DEFINITION 6.5. Average Value If f is integrable on $[a, b]$, the *average value* (or mean value) of f on $[a, b]$ is:

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x)\,dx$$

THEOREM 6.4. Mean-Value Theorem for Integrals If f is continuous on $[a, b]$, there exists a point $c \in [a, b]$ such that:

$$\int_a^b f(x)\,dx = f(c)(b-a)$$

Equivalently, $f(c) = \bar{f}$.

Proof. Since f is continuous on $[a, b]$, it attains a minimum m and a maximum M on $[a, b]$. We have:

$$m(b-a) \leq \int_a^b f(x)\,dx \leq M(b-a)$$

Dividing by $b - a$: $m \leq \bar{f} \leq M$. By the intermediate value theorem for continuous functions, there exists $c \in [a, b]$ with $f(c) = \bar{f}$. □

6.7 The Integral as a Function of the Upper Limit

DEFINITION 6.6. Indefinite Integral If f is integrable on $[a, b]$, the function F defined by:

$$F(x) = \int_a^x f(t)\,dt$$

for $x \in [a, b]$ is called an *indefinite integral* of f.

THEOREM 6.5. If f is integrable on $[a, b]$ and $c \in [a, b]$, then:

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

Proof. This follows directly from the additivity property of the integral. □

THEOREM 6.6. If f is integrable on $[a, b]$ and $F(x) = \int_a^x f(t)\,dt$, then F is continuous on $[a, b]$.

Proof. Since f is integrable, it is bounded, say $|f(x)| \leq M$ for all $x \in [a, b]$. For any $x, y \in [a, b]$ with $x < y$:

$$|F(y) - F(x)| = \left|\int_x^y f(t)\,dt\right| \leq \int_x^y |f(t)|\,dt \leq M(y - x)$$

Given $\epsilon > 0$, choose $\delta = \epsilon/M$. Then $|y - x| < \delta$ implies $|F(y) - F(x)| < \epsilon$. Thus F is continuous (in fact, uniformly continuous) on $[a, b]$. □

7

CONTINUOUS FUNCTIONS

7.1 The Definition of the Limit of a Function

DEFINITION 7.1. Limit of a Function Let f be a function defined on some neighborhood of a point p, although we do not assume that f is defined at the point p itself. Let A be a real number. The statement

$$\lim_{x \to p} f(x) = A$$

means that for every neighborhood $N_\epsilon(A)$ there is a neighborhood $N_\delta(p)$ such that

$$f(x) \in N_\epsilon(A) \quad \text{whenever} \quad x \in N_\delta(p) \quad \text{and} \quad x \neq p$$

In other words, for every $\epsilon > 0$ there is a $\delta > 0$ such that:

$$|f(x) - A| < \epsilon \quad \text{whenever} \quad 0 < |x - p| < \delta$$

DEFINITION 7.2. One-Sided Limits If f is defined on some open interval (p, b), the statement

$$\lim_{x \to p+} f(x) = A$$

means that for every $\epsilon > 0$ there is a $\delta > 0$ such that $|f(x) - A| < \epsilon$ whenever $p < x < p + \delta$. This is called the *right-hand limit.* Similarly, the *left-hand limit* $\lim_{x \to p-} f(x) = A$ is defined using $p - \delta < x < p$.

THEOREM 7.1. Uniqueness of Limits If $\lim_{x \to p} f(x) = A$ and $\lim_{x \to p} f(x) = B$, then $A = B$.

Proof. Suppose $A \neq B$. Let $\epsilon = |A - B|/2$. By the definition of limit, there exists $\delta_1 > 0$ such that $|f(x) - A| < \epsilon$ when $0 < |x - p| < \delta_1$, and $\delta_2 > 0$ such that $|f(x) - B| < \epsilon$ when $0 < |x - p| < \delta_2$. Let $\delta = \min(\delta_1, \delta_2)$. Then for $0 < |x - p| < \delta$, we have both $|f(x) - A| < \epsilon$ and $|f(x) - B| < \epsilon$. By the triangle inequality, $|A - B| \leq |A - f(x)| + |f(x) - B| < 2\epsilon = |A - B|$, a contradiction. $\square$

7.2 The Definition of Continuity of a Function

DEFINITION 7.3. Continuity at a Point A function f is said to be *continuous* at a point p if:

(a) f is defined at p

(b) $\lim_{x \to p} f(x)$ exists

(c) $\lim_{x \to p} f(x) = f(p)$

DEFINITION 7.4. Continuity on an Interval A function f is said to be continuous on an interval I if f is continuous at every point of I.
DEFINITION 7.5. Neighborhood A *neighborhood* of a point p is an open interval containing p. Denoted by $N(p)$ or $N_\delta(p) = (p - \delta, p + \delta)$.

7.3 The Basic Limit Theorems

THEOREM 7.2. Basic Limit Theorems Let f and g be functions such that $\lim_{x\to p} f(x) = A$ and $\lim_{x\to p} g(x) = B$. Then:

(a) $\lim_{x\to p}[f(x) + g(x)] = A + B$

(b) $\lim_{x\to p}[f(x) - g(x)] = A - B$

(c) $\lim_{x\to p} f(x)g(x) = AB$

(d) $\lim_{x\to p} f(x)/g(x) = A/B$, provided $B \neq 0$

(e) $\lim_{x\to p} cf(x) = cA$ for any constant c

Proof. We prove (a) and (c); the others follow similarly.

(a) Given $\epsilon > 0$. There exists $\delta_1 > 0$ such that $|f(x) - A| < \epsilon/2$ when $0 < |x - p| < \delta_1$, and $\delta_2 > 0$ such that $|g(x) - B| < \epsilon/2$ when $0 < |x - p| < \delta_2$. Let $\delta = \min(\delta_1, \delta_2)$. Then for $0 < |x - p| < \delta$:

$$|(f(x) + g(x)) - (A + B)| \leq |f(x) - A| + |g(x) - B| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

(c) Given $\epsilon > 0$. Choose $\delta_1 > 0$ such that $|f(x) - A| < \epsilon/(2|B| + 1)$ when $0 < |x - p| < \delta_1$. Choose $\delta_2 > 0$ such that $|g(x) - B| < \epsilon/(2|A| + 1)$ when $0 < |x - p| < \delta_2$. Choose $\delta_3 > 0$ such that $|g(x) - B| < 1$ (so $|g(x)| < |B| + 1$) when $0 < |x - p| < \delta_3$. Let $\delta = \min(\delta_1, \delta_2, \delta_3)$. Then:

$$|f(x)g(x) - AB| = |f(x)g(x) - Ag(x) + Ag(x) - AB| \leq |g(x)||f(x) - A| + |A||g(x) - B| < (|B| + 1)\frac{\epsilon}{2|B| + 1} +$$

□

THEOREM 7.3. Comparison Theorem for Limits Suppose $f(x) \leq g(x) \leq h(x)$ for all x in some neighborhood of p (except possibly at p itself). If $\lim_{x\to p} f(x) = \lim_{x\to p} h(x) = L$, then $\lim_{x\to p} g(x) = L$.

Proof. Given $\epsilon > 0$. There exists $\delta_1 > 0$ such that $L - \epsilon < f(x) < L + \epsilon$ when $0 < |x - p| < \delta_1$, and $\delta_2 > 0$ such that $L - \epsilon < h(x) < L + \epsilon$ when $0 < |x - p| < \delta_2$. Let $\delta = \min(\delta_1, \delta_2)$. Then for $0 < |x - p| < \delta$:

$$L - \epsilon < f(x) \leq g(x) \leq h(x) < L + \epsilon$$

Thus $|g(x) - L| < \epsilon$. □

7.4 Composite Functions and Continuity

DEFINITION 7.6. Composite Function If u and v are functions, the *composition* of u and v, denoted by $u \circ v$, is the function defined by:

$$(u \circ v)(x) = u(v(x))$$

The domain of $u \circ v$ consists of all x in the domain of v such that $v(x)$ is in the domain of u.

THEOREM 7.4. Continuity of Composite Functions If v is continuous at p and u is continuous at $v(p)$, then the composite function $f = u \circ v$ is continuous at p.

Proof. Given $\epsilon > 0$. Since u is continuous at $v(p)$, there exists $\delta_1 > 0$ such that $|u(y) - u(v(p))| < \epsilon$ whenever $|y - v(p)| < \delta_1$. Since v is continuous at p, there exists $\delta > 0$ such that $|v(x) - v(p)| < \delta_1$ whenever $|x - p| < \delta$. Therefore, $|f(x) - f(p)| = |u(v(x)) - u(v(p))| < \epsilon$ whenever $|x - p| < \delta$. □

7.5 Bolzano's Theorem for Continuous Functions

THEOREM 7.5. Bolzano's Theorem Let f be continuous at each point of a closed interval $[a, b]$ and suppose $f(a)$ and $f(b)$ have opposite signs. Then there is at least one c in the open interval (a, b) such that $f(c) = 0$.

Proof. Assume $f(a) < 0$ and $f(b) > 0$ (the other case is similar). Let $S = \{x \in [a, b] \mid f(x) \leq 0\}$. Then S is nonempty (contains a) and bounded above by b. By the completeness axiom, S has a supremum, call it c. We claim $f(c) = 0$.

Suppose $f(c) > 0$. By continuity, there exists $\delta > 0$ such that $f(x) > 0$ for all $x \in (c - \delta, c + \delta) \cap [a, b]$. Then $c - \delta$ is an upper bound for S, contradicting that c is the least upper bound.

Suppose $f(c) < 0$. By continuity, there exists $\delta > 0$ such that $f(x) < 0$ for all $x \in (c - \delta, c + \delta) \cap [a, b]$. Then there are points in S greater than c, contradicting that c is an upper bound.

Therefore $f(c) = 0$. □

THEOREM 7.6. Intermediate-Value Theorem for Continuous Functions Let f be continuous at each point of a closed interval $[a, b]$. Then f takes on every value between $f(a)$ and $f(b)$ somewhere in the interval.

Proof. Let k be any value between $f(a)$ and $f(b)$. Define $g(x) = f(x) - k$. Then g is continuous on $[a, b]$ and $g(a)$ and $g(b)$ have opposite signs. By Bolzano's theorem, there exists $c \in (a, b)$ such that $g(c) = 0$, i.e., $f(c) = k$. □

7.6 The Process of Inversion

DEFINITION 7.7. Strictly Monotonic Function A function f is said to be *strictly increasing* on an interval I if $f(x) < f(y)$ whenever $x < y$ in I. It is *strictly decreasing* if $f(x) > f(y)$ whenever $x < y$ in I. A function is *strictly monotonic* if it is either strictly increasing or strictly decreasing.

THEOREM 7.7. Existence of Inverse Functions Let f be strictly increasing and continuous on an interval $[a, b]$. Let $c = f(a)$ and $d = f(b)$. Then:

(a) For every y in $[c, d]$ there is exactly one x in $[a, b]$ such that $y = f(x)$.

(b) The inverse function g defined by $g(y) = x$ is strictly increasing and continuous on $[c, d]$.

Proof. (a) Existence follows from the intermediate-value theorem. Uniqueness follows from strict monotonicity: if $f(x_1) = f(x_2) = y$ with $x_1 \neq x_2$, then f is not strictly monotonic.

(b) To show g is strictly increasing: if $y_1 < y_2$, let $x_1 = g(y_1)$ and $x_2 = g(y_2)$. If $x_1 \geq x_2$, then $f(x_1) \geq f(x_2)$ (since f is increasing), so $y_1 \geq y_2$, contradiction.

To show g is continuous: Given $y_0 \in [c, d]$ and $\epsilon > 0$. Let $x_0 = g(y_0)$. Choose δ' such that $[x_0 - \delta', x_0 + \delta'] \subseteq [a, b]$. Since f is continuous and strictly increasing, f maps $[x_0 - \delta', x_0 + \delta']$ onto $[f(x_0 - \delta'), f(x_0 + \delta')]$. Let $\delta = \min(y_0 - f(x_0 - \delta'), f(x_0 + \delta') - y_0)$. Then $|y - y_0| < \delta$ implies $|g(y) - g(y_0)| < \delta'$. □

7.7 The Extreme-Value Theorem for Continuous Functions

THEOREM 7.8. Extreme-Value Theorem If f is continuous on a closed interval $[a, b]$, then there exist points c and d in $[a, b]$ such that:

$$f(c) \leq f(x) \leq f(d) \quad \text{for all } x \in [a, b]$$

Proof. We prove the existence of the maximum; the minimum is similar.

First, f is bounded above on $[a, b]$. If not, for each n there exists $x_n \in [a, b]$ with $f(x_n) > n$. The sequence $\{x_n\}$ has a convergent subsequence by Bolzano-Weierstrass, converging to some $x \in [a, b]$. By continuity, $f(x_n) \to f(x)$, contradicting $f(x_n) > n$.

Let $M = \sup\{f(x) \mid x \in [a, b]\}$. For each n, choose x_n such that $f(x_n) > M - 1/n$. The sequence $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\}$ converging to some $d \in [a, b]$. By continuity, $f(x_{n_k}) \to f(d)$. Since $M - 1/n_k < f(x_{n_k}) \leq M$, we have $f(d) = M$. □

7.8 The Small-Span Theorem for Continuous Functions (Uniform Continuity)

DEFINITION 7.8. Uniform Continuity A function f is said to be *uniformly continuous* on an interval I if for every $\epsilon > 0$ there exists $\delta > 0$ such that $|f(x) - f(y)| < \epsilon$ whenever $x, y \in I$ and $|x - y| < \delta$.

THEOREM 7.9. Small-Span Theorem (Uniform Continuity Theorem) If f is continuous on a closed interval $[a, b]$, then f is uniformly continuous on $[a, b]$.

Proof. Suppose f is not uniformly continuous on $[a, b]$. Then there exists $\epsilon > 0$ such that for every $\delta > 0$, there exist $x, y \in [a, b]$ with $|x - y| < \delta$ but $|f(x) - f(y)| \geq \epsilon$.

Taking $\delta = 1/n$, we obtain sequences $\{x_n\}$ and $\{y_n\}$ in $[a, b]$ such that $|x_n - y_n| < 1/n$ but $|f(x_n) - f(y_n)| \geq \epsilon$. By Bolzano-Weierstrass, $\{x_n\}$ has a convergent subsequence $\{x_{n_k}\}$ converging to some $p \in [a, b]$. Since $|x_{n_k} - y_{n_k}| < 1/n_k$, we also have $y_{n_k} \to p$. By continuity of f:

$$\lim_{k\to\infty} f(x_{n_k}) = f(p) = \lim_{k\to\infty} f(y_{n_k})$$

This contradicts $|f(x_{n_k}) - f(y_{n_k})| \geq \epsilon$ for all k. □

7.9 The Integrability Theorem for Continuous Functions

THEOREM 7.10. Integrability of Continuous Functions If f is continuous on $[a, b]$, then f is integrable on $[a, b]$.

Proof. Since f is continuous on the closed interval $[a, b]$, it is bounded and uniformly continuous.

Given $\epsilon > 0$, by uniform continuity there exists $\delta > 0$ such that $|f(x)-f(y)| < \epsilon/(b-a)$ whenever $|x - y| < \delta$. Choose a partition $P = \{x_0, x_1, \ldots, x_n\}$ with $\Delta x_k < \delta$ for all k.

On each subinterval $[x_{k-1}, x_k]$, let $m_k = \inf f$ and $M_k = \sup f$. By continuity, f attains these values, so there exist $c_k, d_k \in [x_{k-1}, x_k]$ with $f(c_k) = m_k$ and $f(d_k) = M_k$. Since $|c_k - d_k| \leq \Delta x_k < \delta$, we have $M_k - m_k = |f(d_k) - f(c_k)| < \epsilon/(b - a)$.

Therefore:

$$\sum_{k=1}^{n}(M_k - m_k)\Delta x_k < \frac{\epsilon}{b-a}\sum_{k=1}^{n}\Delta x_k = \epsilon$$

This shows the upper and lower integrals differ by less than ϵ, so f is integrable. □

7.10 Mean-Value Theorems for Integrals of Continuous Functions

THEOREM 7.11. First Mean-Value Theorem for Integrals If f is continuous on $[a, b]$, then for some $c \in [a, b]$:

$$\int_a^b f(x)\,dx = f(c)(b - a)$$

Proof. By the extreme-value theorem, f attains minimum m and maximum M on $[a, b]$. Then:

$$m(b - a) \leq \int_a^b f(x)\,dx \leq M(b - a)$$

So $\frac{1}{b-a}\int_a^b f(x)\,dx$ is a value between m and M. By the intermediate-value theorem, there exists $c \in [a, b]$ such that $f(c) = \frac{1}{b-a}\int_a^b f(x)\,dx$. □

THEOREM 7.12. Weighted Mean-Value Theorem for Integrals If f and g are continuous on $[a, b]$, and if g never changes sign on $[a, b]$, then for some $c \in [a, b]$:

$$\int_a^b f(x)g(x)\,dx = f(c)\int_a^b g(x)\,dx$$

Proof. Assume $g(x) \geq 0$ on $[a, b]$. Let $m = \min f$ and $M = \max f$ on $[a, b]$. Then:

$$mg(x) \leq f(x)g(x) \leq Mg(x) \quad \text{for all } x \in [a, b]$$

Integrating: $m\int_a^b g \leq \int_a^b fg \leq M\int_a^b g$. If $\int_a^b g = 0$, the result is trivial. Otherwise, $\frac{\int_a^b fg}{\int_a^b g}$ is a value between m and M, so by the intermediate-value theorem, it equals $f(c)$ for some $c \in [a, b]$. □

Part III

Differential Calculus

8

DIFFERENTIAL CALCULUS

8.1 The Derivative of a Function

DEFINITION 8.1. Derivative Let f be a function defined at least on some open interval containing the point p. The *derivative* of f at p, denoted by $f'(p)$, is defined by the limit:

$$f'(p) = \lim_{h \to 0} \frac{f(p+h) - f(p)}{h}$$

whenever this limit exists. The process of forming this limit is called *differentiation*.

DEFINITION 8.2. Differentiability A function f is said to be *differentiable* at a point p if the derivative $f'(p)$ exists. If f is differentiable at every point of an interval I, we say f is differentiable on I.

THEOREM 8.1. Differentiability Implies Continuity If f is differentiable at p, then f is continuous at p.

Proof. We need to show $\lim_{x \to p} f(x) = f(p)$. For $x \neq p$:

$$f(x) - f(p) = \frac{f(x) - f(p)}{x - p} \cdot (x - p)$$

Taking limits as $x \to p$:

$$\lim_{x \to p} [f(x) - f(p)] = f'(p) \cdot 0 = 0$$

Thus $\lim_{x \to p} f(x) = f(p)$. □

8.2 The Algebra of Derivatives

THEOREM 8.2. Sum Rule If f and g are differentiable at p, then $f + g$ is differentiable at p and:

$$(f + g)'(p) = f'(p) + g'(p)$$

Proof. Using the limit definition:

$$\lim_{h \to 0} \frac{(f+g)(p+h) - (f+g)(p)}{h} = \lim_{h \to 0} \left[\frac{f(p+h) - f(p)}{h} + \frac{g(p+h) - g(p)}{h} \right] = f'(p) + g'(p)$$

□

THEOREM 8.3. Constant Multiple Rule If f is differentiable at p and c is a constant, then cf is differentiable at p and:

$$(cf)'(p) = c \cdot f'(p)$$

Proof. $\lim_{h \to 0} \frac{cf(p+h) - cf(p)}{h} = c \cdot \lim_{h \to 0} \frac{f(p+h) - f(p)}{h} = c \cdot f'(p)$. □

THEOREM 8.4. Product Rule If f and g are differentiable at p, then fg is differentiable at p and:

$$(fg)'(p) = f'(p)g(p) + f(p)g'(p)$$

Proof. We add and subtract $f(p)g(p+h)$:

$$\begin{aligned}\frac{(fg)(p+h) - (fg)(p)}{h} &= \frac{f(p+h)g(p+h) - f(p)g(p)}{h} \\ &= \frac{f(p+h)g(p+h) - f(p)g(p+h) + f(p)g(p+h) - f(p)g(p)}{h} \\ &= g(p+h)\frac{f(p+h) - f(p)}{h} + f(p)\frac{g(p+h) - g(p)}{h}\end{aligned}$$

As $h \to 0$: $g(p+h) \to g(p)$ by continuity, so the limit is $g(p)f'(p) + f(p)g'(p)$. □

THEOREM 8.5. Quotient Rule If f and g are differentiable at p and $g(p) \neq 0$, then f/g is differentiable at p and:

$$\left(\frac{f}{g}\right)'(p) = \frac{f'(p)g(p) - f(p)g'(p)}{[g(p)]^2}$$

Proof. Using the product rule on $f \cdot (1/g)$, we first show $\left(\frac{1}{g}\right)'(p) = -\frac{g'(p)}{[g(p)]^2}$:

$$\lim_{h\to 0} \frac{\frac{1}{g(p+h)} - \frac{1}{g(p)}}{h} = \lim_{h\to 0} \frac{g(p) - g(p+h)}{h \cdot g(p+h)g(p)} = -\frac{g'(p)}{[g(p)]^2}$$

Then apply the product rule to $f \cdot (1/g)$. □

8.3 Derivatives of Standard Functions

THEOREM 8.6. Derivative of a Constant If $f(x) = c$ for all x, then $f'(x) = 0$ for all x.

Proof. $\frac{f(p+h)-f(p)}{h} = \frac{c-c}{h} = 0$, so $f'(p) = \lim_{h\to 0} 0 = 0$. □

THEOREM 8.7. Derivative of the Identity Function If $f(x) = x$ for all x, then $f'(x) = 1$ for all x.

Proof. $\frac{f(p+h)-f(p)}{h} = \frac{(p+h)-p}{h} = \frac{h}{h} = 1$, so $f'(p) = 1$. □

THEOREM 8.8. Power Rule for Positive Integers If n is a positive integer and $f(x) = x^n$, then $f'(x) = nx^{n-1}$.

Proof. By the binomial theorem:

$$\frac{(p+h)^n - p^n}{h} = \frac{np^{n-1}h + \binom{n}{2}p^{n-2}h^2 + \cdots + h^n}{h} = np^{n-1} + \binom{n}{2}p^{n-2}h + \cdots$$

As $h \to 0$, all terms except the first approach 0, giving np^{n-1}. □

THEOREM 8.9. Derivatives of Sine and Cosine

$$\frac{d}{dx}(\sin x) = \cos x \quad \text{and} \quad \frac{d}{dx}(\cos x) = -\sin x$$

Proof. Using the trigonometric identity $\sin(p+h) = \sin p \cos h + \cos p \sin h$:

$$\frac{\sin(p+h) - \sin p}{h} = \sin p \cdot \frac{\cos h - 1}{h} + \cos p \cdot \frac{\sin h}{h}$$

Using the limits $\lim_{h\to 0} \frac{\sin h}{h} = 1$ and $\lim_{h\to 0} \frac{\cos h - 1}{h} = 0$, we get $\cos p$. Similarly for cosine. □

8.4 The Chain Rule

THEOREM 8.10. Chain Rule If f is differentiable at p and g is differentiable at $f(p)$, then the composite function $g \circ f$ is differentiable at p and:

$$(g \circ f)'(p) = g'(f(p)) \cdot f'(p)$$

Proof. Define a function d by:

$$d(y) = \begin{cases} \frac{g(y)-g(f(p))}{y-f(p)} & \text{if } y \neq f(p) \\ g'(f(p)) & \text{if } y = f(p) \end{cases}$$

Then d is continuous at $f(p)$ and $g(y) - g(f(p)) = d(y)(y - f(p))$ for all y. Setting $y = f(p+h)$:

$$\frac{g(f(p+h)) - g(f(p))}{h} = d(f(p+h)) \cdot \frac{f(p+h) - f(p)}{h}$$

As $h \to 0$: $f(p+h) \to f(p)$ by continuity of f, so $d(f(p+h)) \to d(f(p)) = g'(f(p))$. The second factor approaches $f'(p)$. Their product is $g'(f(p)) \cdot f'(p)$. □

8.5 The Mean-Value Theorem for Derivatives

THEOREM 8.11. Rolle's Theorem If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.

Proof. By the extreme-value theorem, f attains maximum M and minimum m on $[a, b]$. If both occur at endpoints, then f is constant and $f'(c) = 0$ for all $c \in (a, b)$. Otherwise, f has an interior extremum at some $c \in (a, b)$. We show $f'(c) = 0$.

For $h > 0$ small enough that $c + h \in (a, b)$: - If $f(c)$ is a maximum: $\frac{f(c+h)-f(c)}{h} \leq 0$, so $f'(c) = \lim_{h\to 0+} \leq 0$ - Also $\frac{f(c-h)-f(c)}{-h} \geq 0$, so $f'(c) = \lim_{h\to 0+} \geq 0$

Thus $f'(c) = 0$. □

THEOREM 8.12. Mean-Value Theorem for Derivatives If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof. Define $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x - a)$. Then $g(a) = f(a)$ and $g(b) = f(b) - (f(b) - f(a)) = f(a) = g(a)$. By Rolle's theorem, there exists $c \in (a, b)$ with $g'(c) = 0$. Since $g'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$, we have $f'(c) = \frac{f(b)-f(a)}{b-a}$. □

THEOREM 8.13. Cauchy's Mean-Value Theorem If f and g are continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that:

$$f'(c)[g(b) - g(a)] = g'(c)[f(b) - f(a)]$$

If $g'(c) \neq 0$ and $g(b) \neq g(a)$, this can be written as:

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Proof. Define $h(x) = f(x)[g(b) - g(a)] - g(x)[f(b) - f(a)]$. Then $h(a) = h(b)$. By Rolle's theorem, there exists $c \in (a, b)$ with $h'(c) = 0$, which gives the result. □

8.6 Applications of Differentiation to Extreme Values

DEFINITION 8.3. Absolute and Relative Extrema A function f has an *absolute maximum* at p if $f(p) \geq f(x)$ for all x in the domain of f. It has a *relative maximum* at p if $f(p) \geq f(x)$ for all x in some neighborhood of p. Similar definitions apply for minima.
DEFINITION 8.4. Critical Point A point p in the domain of f is called a *critical point* if either $f'(p) = 0$ or f is not differentiable at p.
THEOREM 8.14. If f has a relative extremum at an interior point p and f is differentiable at p, then $f'(p) = 0$.

Proof. If f has a relative maximum at p, then for h sufficiently small: - For $h > 0$: $\frac{f(p+h)-f(p)}{h} \leq 0$, so $f'(p) \leq 0$ - For $h < 0$: $\frac{f(p+h)-f(p)}{h} \geq 0$, so $f'(p) \geq 0$ Thus $f'(p) = 0$. □

8.7 Geometric Properties of Functions

THEOREM 8.15. If f is continuous on $[a, b]$ and $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on $[a, b]$.

Proof. For any $x_1, x_2 \in [a, b]$ with $x_1 < x_2$, by the mean-value theorem there exists $c \in (x_1, x_2)$ with $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$. Thus $f(x_1) = f(x_2)$. □
THEOREM 8.16. Increasing Function Theorem If f is continuous on $[a, b]$ and $f'(x) > 0$ for all $x \in (a, b)$, then f is strictly increasing on $[a, b]$.

Proof. For $a \leq x_1 < x_2 \leq b$, by the mean-value theorem there exists $c \in (x_1, x_2)$ with $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) > 0$. Thus $f(x_2) > f(x_1)$. □
THEOREM 8.17. Concavity and the Second Derivative If $f''(x) > 0$ for all $x \in (a, b)$, then f is concave upward on (a, b). If $f''(x) < 0$ for all $x \in (a, b)$, then f is concave downward on (a, b).

Proof. If $f'' > 0$, then f' is increasing. For any $c \in (a, b)$, the tangent line at c has slope $f'(c)$. Since f' is increasing, $f'(x) < f'(c)$ for $x < c$ and $f'(x) > f'(c)$ for $x > c$. This means the graph lies above its tangent lines, i.e., f is concave upward. □
THEOREM 8.18. Second-Derivative Test for Extrema If $f'(p) = 0$ and $f''(p) > 0$, then f has a relative minimum at p. If $f'(p) = 0$ and $f''(p) < 0$, then f has a relative maximum at p.

Proof. Suppose $f'(p) = 0$ and $f''(p) > 0$. Since $f''(p) = \lim_{h\to 0} \frac{f'(p+h)-f'(p)}{h} = \lim_{h\to 0} \frac{f'(p+h)}{h} > 0$, we have $f'(p + h)/h > 0$ for small h. This means $f'(p + h) < 0$ for $h < 0$ and $f'(p + h) > 0$ for $h > 0$. So f decreases to the left of p and increases to the right, giving a minimum. □

8.8 Partial Derivatives

DEFINITION 8.5. Partial Derivative Let f be a function of two variables. The *partial derivative* of f with respect to x at (a, b) is:

$$\frac{\partial f}{\partial x}(a, b) = \lim_{h\to 0} \frac{f(a + h, b) - f(a, b)}{h}$$

whenever this limit exists. Similarly, the partial derivative with respect to y is:

$$\frac{\partial f}{\partial y}(a, b) = \lim_{h\to 0} \frac{f(a, b + h) - f(a, b)}{h}$$

THEOREM 8.19. Equality of Mixed Partial Derivatives If f_{xy} and f_{yx} are both continuous at a point (a, b), then:

$$f_{xy}(a, b) = f_{yx}(a, b)$$

Proof. For small $h, k > 0$, define $\Delta = f(a+h, b+k) - f(a+h, b) - f(a, b+k) + f(a, b)$. Let $g(x) = f(x, b+k) - f(x, b)$. Then $\Delta = g(a+h) - g(a)$. By the mean-value theorem, $\Delta = g'(c)h = [f_x(c, b+k) - f_x(c, b)]h$ for some $c \in (a, a+h)$. Applying the mean-value theorem again to f_x in the second variable: $\Delta = f_{xy}(c, d)hk$ for some $d \in (b, b+k)$.

Similarly, defining $\tilde{g}(y) = f(a+h, y) - f(a, y)$, we get $\Delta = f_{yx}(\tilde{c}, \tilde{d})kh$. Thus $f_{xy}(c, d) = f_{yx}(\tilde{c}, \tilde{d})$. As $h, k \to 0$, continuity gives $f_{xy}(a, b) = f_{yx}(a, b)$. $\square$

9

THE RELATION BETWEEN INTEGRATION AND DIFFERENTIATION

9.1 The First Fundamental Theorem of Calculus

THEOREM 9.1. First Fundamental Theorem of Calculus Let f be a function that is integrable on $[a, b]$ and continuous at a point c in (a, b). Define $A(x) = \int_a^x f(t)\,dt$ for $x \in [a, b]$. Then A is differentiable at c and:

$$A'(c) = f(c)$$

Proof. We need to show that $\lim_{h\to 0} \frac{A(c+h)-A(c)}{h} = f(c)$. For $h > 0$:

$$\frac{A(c+h) - A(c)}{h} = \frac{1}{h}\int_c^{c+h} f(t)\,dt$$

By the mean-value theorem for integrals, there exists $c_h \in [c, c+h]$ such that:

$$\int_c^{c+h} f(t)\,dt = f(c_h) \cdot h$$

Therefore $\frac{A(c+h)-A(c)}{h} = f(c_h)$. As $h \to 0^+$, we have $c_h \to c$, and by continuity of f at c, $f(c_h) \to f(c)$. The same argument applies for $h < 0$. Thus $A'(c) = f(c)$. □

9.2 The Zero-Derivative Theorem

THEOREM 9.2. Zero-Derivative Theorem If f is differentiable on an open interval I and $f'(x) = 0$ for all $x \in I$, then f is constant on I.

Proof. Let $a, b \in I$ with $a < b$. By the mean-value theorem, there exists $c \in (a, b)$ such that $f(b) - f(a) = f'(c)(b - a) = 0$. Thus $f(a) = f(b)$ for all $a, b \in I$, so f is constant. □

9.3 Primitive Functions and the Second Fundamental Theorem

DEFINITION 9.1. Primitive Function A function P is called a *primitive* (or *antiderivative*) of f on an interval I if P is differentiable on I and $P'(x) = f(x)$ for all $x \in I$.

THEOREM 9.3. If P is a primitive of f on an interval I, then every primitive of f on I has the form $P + C$ for some constant C.

Proof. If Q is another primitive of f, then $(Q - P)' = Q' - P' = f - f = 0$. By the zero-derivative theorem, $Q - P = C$ for some constant C. □

THEOREM 9.4. Second Fundamental Theorem of Calculus Let f be continuous on $[a, b]$ and let P be any primitive of f on $[a, b]$. Then:

$$\int_a^b f(x)\,dx = P(b) - P(a)$$

Proof. Define $A(x) = \int_a^x f(t)\,dt$. By the first fundamental theorem, $A'(x) = f(x)$ for all $x \in (a, b)$. Thus A is a primitive of f. Since P is also a primitive, $P(x) = A(x) + C$ for some constant C. Then:

$$P(b) - P(a) = [A(b) + C] - [A(a) + C] = A(b) - A(a) = \int_a^b f(t)\,dt - 0 = \int_a^b f(x)\,dx$$

□

9.4 Integration Formulas

THEOREM 9.5. Integration of Rational Powers For any rational number $r \neq -1$ and any interval where x^r is defined:

$$\int x^r\,dx = \frac{x^{r+1}}{r+1} + C$$

Proof. By differentiation: $\frac{d}{dx}\left(\frac{x^{r+1}}{r+1}\right) = x^r$. □

THEOREM 9.6. Integration of Trigonometric Functions

$$\int \sin x\,dx = -\cos x + C \quad \text{and} \quad \int \cos x\,dx = \sin x + C$$

Proof. By differentiation: $\frac{d}{dx}(-\cos x) = \sin x$ and $\frac{d}{dx}(\sin x) = \cos x$. □

9.5 Integration by Substitution

THEOREM 9.7. Integration by Substitution Let g have a continuous derivative g' on an open interval I. Let $J = g(I)$ and assume f is continuous on J. Then for any $a, b \in I$:

$$\int_a^b f[g(x)]g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$$

Proof. Let P be a primitive of f on J, so $P'(u) = f(u)$. Define $Q(x) = P(g(x))$. By the chain rule:

$$Q'(x) = P'(g(x))g'(x) = f(g(x))g'(x)$$

So Q is a primitive of $f(g(x))g'(x)$. By the second fundamental theorem:

$$\int_a^b f[g(x)]g'(x)\,dx = Q(b) - Q(a) = P(g(b)) - P(g(a)) = \int_{g(a)}^{g(b)} f(u)\,du$$

□

9.6 Integration by Parts

THEOREM 9.8. Integration by Parts If u and v have continuous derivatives on an interval $[a, b]$, then:

$$\int_a^b u(x)v'(x)\,dx = u(b)v(b) - u(a)v(a) - \int_a^b v(x)u'(x)\,dx$$

In differential form, with $du = u'(x)dx$ and $dv = v'(x)dx$:

$$\int u\,dv = uv - \int v\,du$$

Proof. From the product rule: $(uv)' = u'v + uv'$. Integrating both sides from a to b:

$$u(b)v(b) - u(a)v(a) = \int_a^b u'(x)v(x)\,dx + \int_a^b u(x)v'(x)\,dx$$

Rearranging gives the integration by parts formula. □

THEOREM 9.9. Second Mean-Value Theorem for Integrals Assume f is continuous on $[a, b]$ and g has a continuous derivative that never changes sign on $[a, b]$. Then for some $c \in [a, b]$:

$$\int_a^b f(x)g(x)\,dx = f(a)\int_a^c g(x)\,dx + f(b)\int_c^b g(x)\,dx$$

Proof. Let $G(x) = \int_a^x g(t)\,dt$. Integration by parts gives:

$$\int_a^b f(x)g(x)\,dx = f(b)G(b) - \int_a^b f'(x)G(x)\,dx$$

By the weighted mean-value theorem, there exists $c \in [a, b]$ such that:

$$\int_a^b f'(x)G(x)\,dx = G(c)\int_a^b f'(x)\,dx = G(c)[f(b) - f(a)]$$

Substituting back:

$$\int_a^b f(x)g(x)\,dx = f(b)G(b) - G(c)[f(b) - f(a)] = f(a)G(c) + f(b)[G(b) - G(c)]$$

Since $G(c) = \int_a^c g$ and $G(b) - G(c) = \int_c^b g$, the result follows. □

10

TRANSCENDENTAL FUNCTIONS

10.1 The Natural Logarithm

DEFINITION 10.1. Natural Logarithm The *natural logarithm* of x, denoted by $\log x$ or $\ln x$, is defined for all $x > 0$ by the integral:

$$\log x = \int_1^x \frac{1}{t}\, dt$$

THEOREM 10.1. Basic Properties of the Logarithm

(a) $\log 1 = 0$

(b) $\log x > 0$ if $x > 1$, and $\log x < 0$ if $0 < x < 1$

(c) $\log(ab) = \log a + \log b$ for all $a, b > 0$ *(functional equation)*

(d) $\log(a/b) = \log a - \log b$ for all $a, b > 0$

(e) $\log(a^r) = r \log a$ for all $a > 0$ and rational r

Proof. (a) $\log 1 = \int_1^1 \frac{1}{t}\, dt = 0$.
(c) For fixed $a > 0$, define $f(x) = \log(ax) - \log a - \log x$. Then:

$$f'(x) = \frac{1}{ax} \cdot a - \frac{1}{x} = \frac{1}{x} - \frac{1}{x} = 0$$

So $f(x) = C$ (constant). Since $f(1) = \log a - \log a - 0 = 0$, we have $C = 0$. Thus $\log(ax) = \log a + \log x$. □

THEOREM 10.2. Derivative and Integral of the Logarithm

$$\frac{d}{dx} \log x = \frac{1}{x} \quad \text{for all } x > 0$$

$$\int \frac{1}{x}\, dx = \log|x| + C \quad \text{for } x \neq 0$$

Proof. By the first fundamental theorem of calculus, since $\log x = \int_1^x \frac{1}{t}\, dt$, we have $\frac{d}{dx} \log x = \frac{1}{x}$. □

10.2 The Exponential Function

DEFINITION 10.2. Exponential Function Since $\log x$ is strictly increasing and continuous on $(0, \infty)$ with range $(-\infty, \infty)$, it has an inverse function defined on all of $\mathbb{R}$. This inverse is called the *exponential function*, denoted by $\exp(x)$ or e^x.

THEOREM 10.3. Basic Properties of the Exponential Function

(a) $\exp(0) = 1$ and $\exp(1) = e$ where $\log e = 1$

(b) $\exp(a + b) = \exp(a) \cdot \exp(b)$ for all $a, b \in \mathbb{R}$

(c) $\exp(a - b) = \exp(a)/\exp(b)$ for all $a, b \in \mathbb{R}$

(d) $\exp(ra) = [\exp(a)]^r$ for all $a \in \mathbb{R}$ and rational r

Proof. These follow from the corresponding properties of the logarithm. For (b): Let $A = \exp(a)$ and $B = \exp(b)$. Then $a = \log A$ and $b = \log B$, so $a + b = \log A + \log B = \log(AB)$. Therefore $\exp(a + b) = AB = \exp(a)\exp(b)$. □

THEOREM 10.4. Derivative and Integral of the Exponential Function

$$\frac{d}{dx}e^x = e^x \quad \text{for all } x \in \mathbb{R}$$

$$\int e^x \, dx = e^x + C$$

Proof. Let $y = e^x$, so $x = \log y$. By the inverse function theorem:

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{\frac{1}{y}} = y = e^x$$

□

10.3 General Powers and Logarithms

DEFINITION 10.3. General Exponential For $a > 0$ and any real x, define:

$$a^x = e^{x \log a}$$

THEOREM 10.5. Derivatives and Integrals with General Bases For $a > 0$ and $a \neq 1$:

$$\frac{d}{dx}a^x = a^x \log a$$

$$\frac{d}{dx}\log_a x = \frac{1}{x \log a}$$

Proof. $\frac{d}{dx}a^x = \frac{d}{dx}e^{x \log a} = e^{x \log a} \cdot \log a = a^x \log a$. □

10.4 The Hyperbolic Functions

DEFINITION 10.4. Hyperbolic Functions For all real x, define:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

THEOREM 10.6. Derivatives of Hyperbolic Functions

$$\frac{d}{dx}\sinh x = \cosh x, \quad \frac{d}{dx}\cosh x = \sinh x, \quad \frac{d}{dx}\tanh x = \operatorname{sech}^2 x$$

Proof. $\frac{d}{dx}\sinh x = \frac{d}{dx}\left(\frac{e^x - e^{-x}}{2}\right) = \frac{e^x + e^{-x}}{2} = \cosh x$. Similarly for the others. □

THEOREM 10.7. Fundamental Identity

$$\cosh^2 x - \sinh^2 x = 1$$

Proof. $\cosh^2 x - \sinh^2 x = \frac{(e^x+e^{-x})^2}{4} - \frac{(e^x-e^{-x})^2}{4} = \frac{4e^x e^{-x}}{4} = 1.$ □

10.5 Inverse Trigonometric Functions

DEFINITION 10.5. Inverse Sine The function $\sin x$ is strictly increasing and continuous on $[-\pi/2, \pi/2]$, mapping this interval onto $[-1, 1]$. Its inverse, denoted by $\arcsin x$ or $\sin^{-1} x$, is defined on $[-1, 1]$ with range $[-\pi/2, \pi/2]$.

DEFINITION 10.6. Inverse Cosine The function $\cos x$ is strictly decreasing and continuous on $[0, \pi]$, mapping this interval onto $[-1, 1]$. Its inverse, denoted by $\arccos x$ or $\cos^{-1} x$, is defined on $[-1, 1]$ with range $[0, \pi]$.

DEFINITION 10.7. Inverse Tangent The function $\tan x$ is strictly increasing and continuous on $(-\pi/2, \pi/2)$, mapping this interval onto $(-\infty, \infty)$. Its inverse, denoted by $\arctan x$ or $\tan^{-1} x$, is defined on all of $\mathbb{R}$ with range $(-\pi/2, \pi/2)$.

THEOREM 10.8. Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}} \quad \text{for } |x| < 1$$

$$\frac{d}{dx}\arctan x = \frac{1}{1+x^2} \quad \text{for all } x$$

Proof. For $\arcsin x$: Let $y = \arcsin x$, so $x = \sin y$. Then:

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$$

where we take the positive square root since $\cos y \geq 0$ for $y \in [-\pi/2, \pi/2]$. □

THEOREM 10.9. Integration Formulas

$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C = -\arccos x + C$$

$$\int \frac{dx}{1+x^2} = \arctan x + C$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin \frac{x}{a} + C \quad (a > 0)$$

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan \frac{x}{a} + C \quad (a > 0)$$

Proof. These follow directly from the derivative formulas. □

10.6 Integration by Partial Fractions

THEOREM 10.10. Partial Fraction Decomposition Let $P(x)/Q(x)$ be a rational function where $\deg P < \deg Q$ and $Q(x)$ factors as:

$$Q(x) = (x - a_1)^{m_1} \cdots (x - a_k)^{m_k} (x^2 + b_1 x + c_1)^{n_1} \cdots (x^2 + b_l x + c_l)^{n_l}$$

Then $P(x)/Q(x)$ can be written as a sum of terms of the form:

$$\frac{A}{(x-a)^r} \quad \text{and} \quad \frac{Bx + C}{(x^2 + bx + c)^s}$$

THEOREM 10.11. Integration of Rational Functions Every rational function can be integrated in closed form using:

(a) Partial fraction decomposition

(b) $\displaystyle\int \frac{dx}{(x-a)^n} = \begin{cases} \frac{(x-a)^{1-n}}{1-n} & n \neq 1 \\ \log|x-a| & n = 1 \end{cases}$

(c) For irreducible quadratics $x^2 + bx + c$, complete the square and use substitution

Proof. After partial fraction decomposition, each term can be integrated using standard techniques. For quadratic factors, completing the square gives $(x+b/2)^2 + (c - b^2/4)$, which leads to arctan or logarithmic forms. $\square$

Part IV

Sequences and Series

11

POLYNOMIAL APPROXIMATIONS

11.1 The Taylor Polynomials Generated by a Function

DEFINITION 11.1. Taylor Polynomial Let f be a function with derivatives of order n at the point a. The polynomial:

$$P_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

is called the nth Taylor polynomial of f at a, or generated by f at a.

Note: **Taylor Operator Notation.** It is convenient to have a notation that indicates the dependence of the Taylor polynomial P on f and n. We shall indicate this dependence by writing $P = T_n f$ or $P = T_n(f)$. The symbol T_n is called the **Taylor operator** of degree n. When this operator is applied to a function f, it produces a new function $T_n f$, the Taylor polynomial of degree n. The value of this function at x is denoted by $T_n f(x)$ or by $T_n[f(x)]$. If we also wish to indicate the dependence on a, we write $T_n f(x; a)$ instead of $T_n f(x)$.

THEOREM 11.1. Approximation Property of Taylor Polynomials Let f have derivatives of order n at the point a, and let P_n be the nth Taylor polynomial of f at a. Then:

$$P_n(a) = f(a), \quad P_n'(a) = f'(a), \quad P_n''(a) = f''(a), \quad \ldots, \quad P_n^{(n)}(a) = f^{(n)}(a)$$

Proof. Differentiating $P_n(x)$ term by term k times $(0 \le k \le n)$ gives:

$$P_n^{(k)}(x) = f^{(k)}(a) + \frac{f^{(k+1)}(a)}{1!}(x-a) + \cdots + \frac{f^{(n)}(a)}{(n-k)!}(x-a)^{n-k}$$

Setting $x = a$, all terms with $(x-a)$ vanish, leaving $P_n^{(k)}(a) = f^{(k)}(a)$. □

THEOREM 11.2. Uniqueness of the Approximating Polynomial Let f have derivatives of order n at a, and let P be a polynomial of degree $\le n$ such that:

$$P(a) = f(a), \quad P'(a) = f'(a), \quad \ldots, \quad P^{(n)}(a) = f^{(n)}(a)$$

Then $P = P_n$, the nth Taylor polynomial of f at a.

Proof. Write $P(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n$. Successive differentiation and evaluation at $x = a$ gives $c_k = f^{(k)}(a)/k!$ for each k, so $P = P_n$. □

11.2 Calculus of Taylor Polynomials

THEOREM 11.3. Linearity Property If P_n and Q_n are the nth Taylor polynomials of f and g at a, then the nth Taylor polynomial of $\alpha f + \beta g$ is $\alpha P_n + \beta Q_n$. In operator notation:

$$T_n(c_1 f + c_2 g) = c_1 T_n(f) + c_2 T_n(g)$$

Proof. By the uniqueness theorem, since $\alpha P_n + \beta Q_n$ matches the derivatives of $\alpha f + \beta g$ up to order n at a, it is the Taylor polynomial. □

THEOREM 11.4. Differentiation Property The derivative of the nth Taylor polynomial of f at a is the $(n-1)$st Taylor polynomial of f' at a. In operator notation:

$$(T_n f)' = T_{n-1}(f')$$

Proof. Differentiating $P_n(x)$ gives:

$$P_n'(x) = f'(a) + \frac{f''(a)}{1!}(x-a) + \cdots + \frac{f^{(n)}(a)}{(n-1)!}(x-a)^{n-1}$$

which is precisely the $(n-1)$st Taylor polynomial of f' at a. □

THEOREM 11.5. Integration Property If f has $n+1$ derivatives at a, the $(n+1)$st Taylor polynomial of $g(x) = \int_a^x f(t)\,dt$ is $\int_a^x T_n f(t)\,dt$. In operator notation:

$$T_{n+1} g(x) = \int_a^x T_n f(t)\,dt$$

Proof. Let $Q(x) = \int_a^x P_n(t)\,dt$. Then $Q(a) = 0$ and $Q'(x) = P_n(x)$. By the differentiation property, $Q^{(k+1)}(a) = P_n^{(k)}(a) = f^{(k)}(a)$ for $0 \le k \le n$, so $Q^{(k+1)}(a) = f^{(k)}(a) = P^{(k+1)}(a)$. Since $P(a) = 0 = Q(a)$, we have $Q^{(k)}(a) = P^{(k)}(a)$ for $0 \le k \le n+1$, proving Q is the Taylor polynomial of P. □

THEOREM 11.6. Substitution Property Let $g(x) = f(cx)$, where c is a constant. Then:

$$T_n g(x; a) = T_n f(cx; ca)$$

In particular, when $a = 0$, we have $T_n g(x) = T_n f(cx)$.

Proof. Since $g(x) = f(cx)$, the chain rule gives $g'(x) = cf'(cx)$, $g''(x) = c^2 f''(cx)$, ..., $g^{(k)}(x) = c^k f^{(k)}(cx)$. Hence:

$$T_n g(x; a) = \sum_{k=0}^{n} \frac{g^{(k)}(a)}{k!}(x-a)^k = \sum_{k=0}^{n} \frac{c^k f^{(k)}(ca)}{k!}(x-a)^k = \sum_{k=0}^{n} \frac{f^{(k)}(ca)}{k!}(cx-ca)^k = T_n f(cx; ca).$$

□

THEOREM 11.7. Characterization of Taylor Polynomials Let P_n be a polynomial of degree $\le n$. Let f and g be two functions with derivatives of order n at 0 and assume that:

$$f(x) = P_n(x) + x^n g(x)$$

where $g(x) \to 0$ as $x \to 0$. Then P_n is the Taylor polynomial generated by f at 0, that is, $P_n = T_n f$.

Proof. Let $h(x) = f(x) - P_n(x) = x^n g(x)$. By differentiating the product $x^n g(x)$ repeatedly, we see that h and its first n derivatives are 0 at $x = 0$. Therefore, f agrees with P_n and its first n derivatives at 0, so $P_n = T_n f$ by the uniqueness theorem. □

11.3 Taylor's Formula with Remainder

We turn now to a discussion of the error in the approximation of a function f by its Taylor polynomial $T_n f$ at a point a. The error is defined to be the difference:

$$E_n(x) = f(x) - T_n f(x)$$

Thus, if f has a derivative of order n at a, we may write:

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + E_n(x)$$

This is known as **Taylor's formula with remainder** $E_n(x)$; it is useful whenever we can estimate the size of $E_n(x)$. We shall express the error as an integral and then estimate the size of the integral. To illustrate the principal ideas, we consider first the error arising from a linear approximation.

THEOREM 11.8. Taylor's Formula with Integral Remainder (First Order) Assume f has a continuous second derivative f'' in some neighborhood of a. Then, for every x in this neighborhood, we have:

$$f(x) = f(a) + f'(a)(x-a) + E_1(x)$$

where:

$$E_1(x) = \int_a^x (x-t)f''(t)\,dt$$

Proof. From the definition of the error we may write:

$$E_1(x) = f(x) - f(a) - f'(a)(x-a) = \int_a^x f'(t)\,dt - f'(a)\int_a^x dt = \int_a^x [f'(t) - f'(a)]\,dt$$

The last integral may be written as $\int_a^x u\,dv$, where $u = f'(t) - f'(a)$ and $v = t - x$. Now $du/dt = f''(t)$ and $dv/dt = 1$, so integration by parts gives:

$$E_1(x) = \int_a^x u\,dv = uv\Big|_a^x - \int_a^x (t-x)f''(t)\,dt = \int_a^x (x-t)f''(t)\,dt$$

since $u = 0$ when $t = a$, and $v = 0$ when $t = x$. □

The corresponding result for a polynomial approximation of degree n is given by the following.

THEOREM 11.9. Taylor's Formula with Integral Remainder Assume f has a continuous derivative of order $n+1$ in some interval containing a. Then, for every x in this interval, we have the Taylor formula:

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + E_n(x)$$

where:

$$E_n(x) = \frac{1}{n!}\int_a^x (x-t)^n f^{(n+1)}(t)\,dt$$

Proof. The theorem is proved by induction on n. We have already proved it for $n = 1$. Now we assume it is true for some n and prove it for $n+1$. We write Taylor's formula with $n+1$ and with n and subtract to get:

$$E_{n+1}(x) = E_n(x) - \frac{f^{(n+1)}(a)}{(n+1)!}(x-a)^{n+1}$$

Now we use the integral for $E_n(x)$ and note that $(x-a)^{n+1}/(n+1) = \int_a^x (x-t)^n\,dt$ to obtain:

$$\begin{aligned} E_{n+1}(x) &= \frac{1}{n!}\int_a^x (x-t)^n f^{(n+1)}(t)\,dt - \frac{f^{(n+1)}(a)}{n!}\int_a^x (x-t)^n\,dt \\ &= \frac{1}{n!}\int_a^x (x-t)^n [f^{(n+1)}(t) - f^{(n+1)}(a)]\,dt \end{aligned}$$

The last integral may be written in the form $\int_a^x u\,dv$, where $u = f^{(n+1)}(t) - f^{(n+1)}(a)$ and $v = -(x-t)^{n+1}/(n+1)$. Integrating by parts and noting that $u = 0$ when $t = a$, and that $v = 0$ when $t = x$, we find that:

$$E_{n+1}(x) = \frac{1}{n!}\int_a^x u\,dv = -\frac{1}{n!}\int_a^x v\,du = \frac{1}{(n+1)!}\int_a^x (x-t)^{n+1} f^{(n+2)}(t)\,dt$$

This completes the inductive step from n to $n+1$, so the theorem is true for all $n \geq 1$. □

11.4 Other Forms of the Remainder in Taylor's Formula

THEOREM 11.10. Taylor's Formula with Lagrange Remainder Assume f has a continuous derivative of order $n+1$ in some interval containing a. Then, for every x in this interval, we have the Taylor formula:

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + E_n(x),$$

where

$$E_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

for some point c between a and x.

Proof. Fix $x \neq a$ and define for t between a and x:

$$F(t) = f(x) - f(t) - f'(t)(x-t) - \frac{f''(t)}{2!}(x-t)^2 - \cdots - \frac{f^{(n)}(t)}{n!}(x-t)^n$$

Then $F(a) = R_n(x)$ and $F(x) = 0$. Differentiating:

$$F'(t) = -\frac{f^{(n+1)}(t)}{n!}(x-t)^n$$

By the mean-value theorem, there exists c between a and x such that:

$$\frac{F(x) - F(a)}{x-a} = F'(c) = -\frac{f^{(n+1)}(c)}{n!}(x-c)^n$$

Since $F(x) - F(a) = -R_n(x)$, we get $R_n(x) = \frac{f^{(n+1)}(c)}{n!}(x-c)^n(x-a)$. A more careful analysis (using Cauchy's mean-value theorem) gives the stated form. □

THEOREM 11.11. Estimates for the Error in Taylor's Formula If the $(n+1)$st derivative of f satisfies the inequalities

$$m \leq f^{(n+1)}(t) \leq M$$

for all t in some interval containing a, then for every x in this interval we have the following estimates:

$$m\frac{(x-a)^{n+1}}{(n+1)!} \leq E_n(x) \leq M\frac{(x-a)^{n+1}}{(n+1)!} \quad \text{if } x > a,$$

and

$$m\frac{(a-x)^{n+1}}{(n+1)!} \leq (-1)^{n+1}E_n(x) \leq M\frac{(a-x)^{n+1}}{(n+1)!} \quad \text{if } x < a.$$

Proof. Assume first that $x > a$. Then the integral for $E_n(x)$ is extended over the interval $[a, x]$. For each t in this interval we have $(x-t)^n \geq 0$, so the inequalities give us

$$m\frac{(x-t)^n}{n!} \leq \frac{(x-t)^n}{n!} f^{(n+1)}(t) \leq M\frac{(x-t)^n}{n!}.$$

Integrating from a to x, we find that

$$\frac{m}{n!}\int_a^x (x-t)^n\,dt \leq E_n(x) \leq \frac{M}{n!}\int_a^x (x-t)^n\,dt.$$

The substitution $u = x - t$, $du = -dt$ gives us

$$\int_a^x (x-t)^n\,dt = \int_0^{x-a} u^n\,du = \frac{(x-a)^{n+1}}{n+1},$$

so the inequalities reduce to the stated form for $x > a$.

If $x < a$, the integration takes place over the interval $[x, a]$. For each t in this interval we have $t \geq x$, so $(-1)^n(x-t)^n = (t-x)^n \geq 0$. Therefore, we may multiply the inequalities by the nonnegative factor $(-1)^n(x-t)^n/n!$ and integrate from x to a to obtain the second set of inequalities. □

THEOREM 11.12. Estimate of the Remainder If f has derivatives of order $n+1$ in an interval I containing a, and if $|f^{(n+1)}(x)| \leq M$ for all $x \in I$, then:

$$|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1}$$

for all $x \in I$.

Proof. From Taylor's formula with $|f^{(n+1)}(c)| \leq M$:

$$|R_n(x)| = \left|\frac{f^{(n+1)}(c)}{(n+1)!}\right| |x-a|^{n+1} \leq \frac{M}{(n+1)!}|x-a|^{n+1}$$

□

THEOREM 11.13. Taylor's Formula with Cauchy Remainder Under the same hypotheses as the integral form, for each $x \in I$, there exists c between a and x such that:

$$R_n(x) = \frac{f^{(n+1)}(c)}{n!}(x-c)^n(x-a)$$

Proof. Apply the weighted mean-value theorem for integrals to the integral remainder with $g(t) = (x-t)^n$. There exists c between a and x such that:

$$\int_a^x (x-t)^n f^{(n+1)}(t)\,dt = f^{(n+1)}(c)(x-c)^n(x-a)$$

Substituting into the integral form gives the result. □

11.5 Error Estimates

THEOREM 11.14. Little-o Notation for Remainder If f has derivatives of order n at a, then:

$$R_n(x) = o(|x-a|^n) \quad \text{as } x \to a$$

That is, $\lim_{x\to a} \frac{R_n(x)}{(x-a)^n} = 0$.

Proof. By Taylor's formula with the remainder estimate, if $f^{(n+1)}$ is bounded near a, then $|R_n(x)| \leq M|x-a|^{n+1}/(n+1)! = o(|x-a|^n)$. □

11.6 Applications to Indeterminate Forms

DEFINITION 11.2. Indeterminate Form 0/0 If $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$, the quotient $f(x)/g(x)$ is said to assume the indeterminate form $0/0$ at $x = a$.

THEOREM 11.15. L'Hôpital's Rule for 0/0 Assume f and g have derivatives f' and g' at each point of an open interval (a, b), and that $g'(x) \neq 0$ for all $x \in (a, b)$. Assume also that:

$$\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0$$

If $\lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L$ (where L may be finite or infinite), then:

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L$$

Proof. Extend f and g to be continuous at a by defining $f(a) = g(a) = 0$. By the generalized mean-value theorem, for each $x \in (a, b)$, there exists $c \in (a, x)$ such that:

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)}$$

As $x \to a^+$, we have $c \to a^+$, so:

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = \lim_{c\to a^+} \frac{f'(c)}{g'(c)} = L$$

□

THEOREM 11.16. L'Hôpital's Rule for ∞/∞ Assume f and g have derivatives on (a, b) with $g'(x) \neq 0$. If $\lim_{x\to a^+} |f(x)| = \lim_{x\to a^+} |g(x)| = \infty$ and $\lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L$, then:

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L$$

Proof. We prove the case where L is finite. Given $\epsilon > 0$, there exists $c \in (a, b)$ such that $\left|\frac{f'(t)}{g'(t)} - L\right| < \epsilon$ for all $t \in (a, c)$. For $a < x < y < c$, by the generalized mean-value theorem, there exists $t \in (x, y)$ with:

$$\frac{f(y) - f(x)}{g(y) - g(x)} = \frac{f'(t)}{g'(t)}$$

Rearranging: $\frac{f(x)}{g(x)} = \frac{f(y)}{g(x)} + \left(1 - \frac{g(y)}{g(x)}\right) \frac{f'(t)}{g'(t)}$.

As $x \to a^+$ with y fixed, since $|g(x)| \to \infty$, we have $\frac{f(y)}{g(x)} \to 0$ and $\frac{g(y)}{g(x)} \to 0$. Thus for x close enough to a:

$$\left|\frac{f(x)}{g(x)} - \frac{f'(t)}{g'(t)}\right| < \epsilon$$

Since $\left|\frac{f'(t)}{g'(t)} - L\right| < \epsilon$, we get $\left|\frac{f(x)}{g(x)} - L\right| < 2\epsilon$. □

11.7 The Symbols ∞ and $-\infty$

DEFINITION 11.3. Limit to Infinity We say $\lim_{x\to a} f(x) = +\infty$ if for every $M > 0$, there exists $\delta > 0$ such that $f(x) > M$ whenever $0 < |x - a| < \delta$.

DEFINITION 11.4. Infinite Limits

$$\lim_{x\to a^-} f(x) = +\infty$$

means that for every $M > 0$, there exists $\delta > 0$ such that $f(x) > M$ whenever $a - \delta < x < a$.

$$\lim_{x\to a^+} f(x) = +\infty$$

means that for every $M > 0$, there exists $\delta > 0$ such that $f(x) > M$ whenever $a < x < a + \delta$.

THEOREM 11.17. Behavior of $\log x$ and e^x for Large x For any $\alpha > 0$:

$$\lim_{x\to+\infty} \frac{\log x}{x^\alpha} = 0$$

$$\lim_{x\to+\infty} \frac{x^\alpha}{e^x} = 0$$

Proof. For the first limit, apply L'Hôpital's rule:

$$\lim_{x\to+\infty} \frac{\log x}{x^\alpha} = \lim_{x\to+\infty} \frac{1/x}{\alpha x^{\alpha-1}} = \lim_{x\to+\infty} \frac{1}{\alpha x^\alpha} = 0$$

For the second limit, apply L'Hôpital's rule n times where $n > \alpha$:

$$\lim_{x\to+\infty} \frac{x^\alpha}{e^x} = \lim_{x\to+\infty} \frac{\alpha x^{\alpha-1}}{e^x} = \cdots = \lim_{x\to+\infty} \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)x^{\alpha-n}}{e^x} = 0$$

since $\alpha - n < 0$. □

THEOREM 11.18. Exponential vs. Polynomial Growth For every $n > 0$:

$$\lim_{x\to+\infty} \frac{x^n}{e^x} = 0$$

$$\lim_{x\to+\infty} \frac{e^x}{x^n} = +\infty$$

Proof. Apply L'Hôpital's rule n times to the first limit; each application reduces the power of x by 1 until the numerator is constant and the denominator is e^x. The second limit follows from the first. □

12

DIFFERENTIAL EQUATIONS

12.1 Terminology and Notation

DEFINITION 12.1. Differential Equation A differential equation is an equation involving an unknown function and one or more of its derivatives. An equation of the form $y' = f(x, y)$ is called a first-order differential equation. An equation of the form $y'' = f(x, y, y')$ is called a second-order differential equation.

DEFINITION 12.2. Solution of a Differential Equation A function $y = Y(x)$ is said to be a solution of the differential equation $y' = f(x, y)$ on an interval I if Y is differentiable on I and satisfies $Y'(x) = f(x, Y(x))$ for all $x \in I$.

12.2 First-Order Linear Differential Equations

DEFINITION 12.3. First-Order Linear Equation A first-order differential equation of the form:

$$y' + P(x)y = Q(x)$$

where P and Q are given functions, is called a first-order linear differential equation.

THEOREM 12.1. General Solution of First-Order Linear Equations Assume P and Q are continuous on an open interval I. Choose any point a in I and let b be any real number. Then there is one and only one function $y = f(x)$ which satisfies the initial-value problem:

$$y' + P(x)y = Q(x), \quad \text{with } f(a) = b$$

on the interval I. This function is given by the formula:

$$f(x) = be^{-A(x)} + e^{-A(x)} \int_a^x Q(t)e^{A(t)}\, dt$$

where $A(x) = \int_a^x P(t)\, dt$.

Proof. First we verify that the function satisfies the differential equation. Let $v(x) = e^{A(x)}$. Then $v'(x) = P(x)e^{A(x)} = P(x)v(x)$. Multiplying the differential equation by $v(x)$:

$$v(x)y' + P(x)v(x)y = v(x)Q(x)$$

Since $v'(x) = P(x)v(x)$, the left side is $(v(x)y)'$. Integrating from a to x:

$$v(x)y(x) - v(a)y(a) = \int_a^x Q(t)v(t)\, dt$$

Since $v(a) = e^{A(a)} = e^0 = 1$ and $y(a) = b$:

$$y(x) = \frac{b + \int_a^x Q(t)v(t)\, dt}{v(x)} = be^{-A(x)} + e^{-A(x)} \int_a^x Q(t)e^{A(t)}\, dt$$

For uniqueness, suppose f and g are both solutions with $f(a) = g(a) = b$. Let $h = f - g$. Then $h(a) = 0$ and $h' + P(x)h = 0$. Multiplying by the integrating factor $e^{A(x)}$:

$$\frac{d}{dx}[h(x)e^{A(x)}] = 0$$

So $h(x)e^{A(x)} = h(a)e^{A(a)} = 0$, implying $h(x) = 0$ for all $x \in I$. Thus $f = g$. □

12.3 Existence and Uniqueness Theorems

THEOREM 12.2. Existence and Uniqueness for $y' = f(x, y)$ Let $f(x, y)$ be continuous and satisfy a Lipschitz condition in y on a rectangle $R = [a, b] \times [c, d]$. That is, there exists $L > 0$ such that:

$$|f(x, y_1) - f(x, y_2)| \leq L|y_1 - y_2|$$

for all $(x, y_1), (x, y_2) \in R$. Then for any (x_0, y_0) in the interior of R, there exists a unique solution to the initial-value problem $y' = f(x, y)$, $y(x_0) = y_0$ on some interval containing x_0.

Proof. (Sketch) The proof uses Picard iteration. Define a sequence of functions recursively:

$$Y_0(x) = y_0, \quad Y_{n+1}(x) = y_0 + \int_{x_0}^{x} f(t, Y_n(t))\, dt$$

One shows this sequence converges uniformly to a solution. The Lipschitz condition ensures the solution is unique by showing any two solutions must coincide. □

12.4 Second-Order Linear Differential Equations with Constant Coefficients

DEFINITION 12.4. Second-Order Linear Equation A differential equation of the form:

$$y'' + ay' + by = R(x)$$

where a and b are constants and R is a given function, is called a second-order linear differential equation with constant coefficients. The equation is homogeneous if $R(x) = 0$.

DEFINITION 12.5. Characteristic Equation For the homogeneous equation $y'' + ay' + by = 0$, the quadratic equation:

$$r^2 + ar + b = 0$$

is called the characteristic equation. Its roots are denoted by r_1 and r_2.

THEOREM 12.3. General Solution of Homogeneous Equation The general solution of $y'' + ay' + by = 0$ depends on the discriminant $D = a^2 - 4b$:

(a) If $D > 0$ (distinct real roots $r_1 \neq r_2$): $y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$

(b) If $D = 0$ (repeated root r): $y = (c_1 + c_2 x)e^{rx}$

(c) If $D < 0$ (complex roots $\alpha \pm i\beta$): $y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$

Proof. Substituting $y = e^{rx}$ into the differential equation gives the characteristic equation.

Case (a): If $r_1 \neq r_2$ are real, both $e^{r_1 x}$ and $e^{r_2 x}$ satisfy the equation. Their Wronskian $W = (r_2 - r_1)e^{(r_1+r_2)x} \neq 0$, so they are linearly independent.

Case (b): If r is a repeated root, e^{rx} and xe^{rx} are solutions. One verifies directly that $y = xe^{rx}$ satisfies the equation.

Case (c): For complex roots $\alpha \pm i\beta$, using Euler's formula $e^{(\alpha+i\beta)x} = e^{\alpha x}(\cos\beta x + i\sin\beta x)$, the real and imaginary parts give real solutions $e^{\alpha x}\cos\beta x$ and $e^{\alpha x}\sin\beta x$. □

THEOREM 12.4. General Solution of Nonhomogeneous Equation If y_p is any particular solution of $y'' + ay' + by = R(x)$, and y_1, y_2 are linearly independent solutions of the homogeneous equation, then every solution has the form:

$$y = c_1 y_1 + c_2 y_2 + y_p$$

where c_1, c_2 are constants.

Proof. Let y be any solution and y_p be a particular solution. Then $(y - y_p)'' + a(y - y_p)' + b(y - y_p) = R - R = 0$. So $y - y_p$ satisfies the homogeneous equation, hence $y - y_p = c_1 y_1 + c_2 y_2$ for some constants c_1, c_2. □

THEOREM 12.5. Variation of Parameters If y_1 and y_2 are linearly independent solutions of $y'' + ay' + by = 0$, then a particular solution of $y'' + ay' + by = R(x)$ is:

$$y_p = -y_1 \int \frac{y_2 R}{W}\,dx + y_2 \int \frac{y_1 R}{W}\,dx$$

where $W = y_1 y_2' - y_2 y_1'$ is the Wronskian.

Proof. Try a solution of the form $y_p = v_1 y_1 + v_2 y_2$ where v_1 and v_2 are functions to be determined. Imposing the condition $v_1' y_1 + v_2' y_2 = 0$, we get:

$$y_p' = v_1 y_1' + v_2 y_2', \quad y_p'' = v_1' y_1' + v_2' y_2' + v_1 y_1'' + v_2 y_2''$$

Substituting into the differential equation and using that y_1, y_2 satisfy the homogeneous equation:

$$v_1' y_1' + v_2' y_2' = R$$

Solving the system:

$$\begin{cases} v_1' y_1 + v_2' y_2 = 0 \\ v_1' y_1' + v_2' y_2' = R \end{cases}$$

gives $v_1' = -\frac{y_2 R}{W}$ and $v_2' = \frac{y_1 R}{W}$. Integrating yields the formula. □

12.5 Separable First-Order Equations

DEFINITION 12.6. Separable Equation A first-order differential equation of the form $y' = f(x, y)$ is called separable if $f(x, y)$ can be written as:

$$f(x, y) = \frac{Q(x)}{P(y)}$$

where P and Q are functions of y and x respectively.

THEOREM 12.6. Solution of Separable Equations If $y' = Q(x)/P(y)$ with $P(y) \neq 0$, and G is any primitive of P (i.e., $G' = P$), then every solution satisfies:

$$G(y) = \int Q(x)\,dx + C$$

for some constant C.

Proof. Write the equation as $P(y)\frac{dy}{dx} = Q(x)$. If $y = Y(x)$ is a solution, then by the chain rule:

$$\frac{d}{dx}G(Y(x)) = P(Y(x))Y'(x) = Q(x)$$

Integrating both sides: $G(Y(x)) = \int Q(x)\,dx + C$. □

12.6 Homogeneous First-Order Equations

DEFINITION 12.7. Homogeneous First-Order Equation A first-order differential equation $y' = f(x, y)$ is called homogeneous (in the sense of differential equations) if $f(tx, ty) = f(x, y)$ for all $t \neq 0$. Equivalently, f can be written as $f(x, y) = g(y/x)$.

THEOREM 12.7. Solution of Homogeneous Equations The homogeneous equation $y' = g(y/x)$ can be transformed into a separable equation by the substitution $v = y/x$ (i.e., $y = vx$). Then:

$$\frac{dv}{g(v) - v} = \frac{dx}{x}$$

Proof. With $y = vx$, we have $y' = v + xv'$. The equation becomes:

$$v + xv' = g(v)$$

Rearranging: $xv' = g(v) - v$, or $\frac{dv}{g(v)-v} = \frac{dx}{x}$. This is separable in v and x. □

13

COMPLEX NUMBERS

13.1 Definition and Field Properties

DEFINITION 13.1. Complex Number A complex number is an ordered pair (a, b) of real numbers. Two complex numbers (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$. The set of all complex numbers is denoted by $\mathbb{C}$.

DEFINITION 13.2. Addition and Multiplication For complex numbers (a, b) and (c, d), define:

$$(a, b) + (c, d) = (a + c, b + d)$$

$$(a, b) \cdot (c, d) = (ac - bd, ad + bc)$$

THEOREM 13.1. Field Properties The set $\mathbb{C}$ with the operations of addition and multiplication forms a field. That is:

(a) Addition is commutative and associative

(b) Multiplication is commutative and associative

(c) Distributive law: $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$

(d) Additive identity: $(0, 0)$ satisfies $(a, b) + (0, 0) = (a, b)$

(e) Multiplicative identity: $(1, 0)$ satisfies $(a, b) \cdot (1, 0) = (a, b)$

(f) Additive inverses exist for all (a, b)

(g) Multiplicative inverses exist for all $(a, b) \neq (0, 0)$

Proof. The algebraic properties follow directly from the definitions and the corresponding properties of real numbers. For multiplicative inverses, if $(a, b) \neq (0, 0)$, then:

$$(a, b)^{-1} = \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right)$$

Verification: $(a, b) \cdot \left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2} \right) = \left(\frac{a^2+b^2}{a^2+b^2}, \frac{-ab+ab}{a^2+b^2} \right) = (1, 0)$. $\square$

13.2 The Imaginary Unit

DEFINITION 13.3. Imaginary Unit The imaginary unit i is defined as the complex number $i = (0, 1)$.

THEOREM 13.2. Basic Property of i

$$i^2 = -1$$

Proof. $i^2 = (0,1) \cdot (0,1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0) = (-1, 0) = -1$. □

THEOREM 13.3. Standard Form Every complex number (a, b) can be written uniquely as:

$$(a, b) = a + bi$$

where a and b are real numbers.

Proof. $(a, b) = (a, 0) + (0, b) = (a, 0) + (b, 0) \cdot (0, 1) = a + bi$, identifying real number a with complex number $(a, 0)$. □

13.3 Geometric Interpretation

DEFINITION 13.4. Modulus and Argument For a complex number $z = x + yi$:

(a) The *modulus* (absolute value) is $|z| = \sqrt{x^2 + y^2}$

(b) The *argument* $\theta = \arg(z)$ satisfies $\cos\theta = x/|z|$ and $\sin\theta = y/|z|$ (for $z \neq 0$)

THEOREM 13.4. Polar Form Every nonzero complex number can be written in polar form:

$$z = r(\cos\theta + i\sin\theta)$$

where $r = |z|$ and $\theta = \arg(z)$.

Proof. If $z = x + yi$, then $x = r\cos\theta$ and $y = r\sin\theta$ where $r = \sqrt{x^2 + y^2}$. Thus $z = r(\cos\theta + i\sin\theta)$. □

THEOREM 13.5. Multiplication in Polar Form If $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$, then:

$$z_1 \cdot z_2 = r_1 r_2[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)]$$

and:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}[\cos(\theta_1 - \theta_2) + i\sin(\theta_1 - \theta_2)]$$

Proof. Using the angle addition formulas:

$$\begin{aligned} z_1 \cdot z_2 &= r_1 r_2[(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)] \\ &= r_1 r_2[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)] \end{aligned}$$

The division formula follows similarly. □

13.4 Exponential Form

DEFINITION 13.5. Complex Exponential For a complex number $z = x + iy$, define:

$$e^z = e^x(\cos y + i\sin y)$$

In particular, for pure imaginary $z = i\theta$:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

THEOREM 13.6. Euler's Formula For any real number θ:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

In particular:

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

Proof. This follows directly from the definition of the complex exponential. The formulas for cosine and sine are obtained by adding and subtracting $e^{i\theta}$ and $e^{-i\theta}$. □

THEOREM 13.7. Properties of the Complex Exponential For all complex numbers z, z_1, z_2:

(a) $e^{z_1+z_2} = e^{z_1} \cdot e^{z_2}$

(b) $e^{z_1-z_2} = e^{z_1}/e^{z_2}$

(c) $e^z \neq 0$ for all $z \in \mathbb{C}$

(d) $|e^z| = e^{\mathrm{Re}(z)}$

(e) $e^{z+2\pi i} = e^z$ (periodic with period $2\pi i$)

Proof. (a) Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$. Then:

$$\begin{aligned} e^{z_1} \cdot e^{z_2} &= e^{x_1}(\cos y_1 + i\sin y_1) \cdot e^{x_2}(\cos y_2 + i\sin y_2) \\ &= e^{x_1+x_2}[\cos(y_1+y_2) + i\sin(y_1+y_2)] \\ &= e^{z_1+z_2} \end{aligned}$$

using the multiplication rule for polar forms.

(d) $|e^z| = |e^x(\cos y + i\sin y)| = e^x \cdot 1 = e^{\mathrm{Re}(z)}$.

(e) $e^{z+2\pi i} = e^x[\cos(y+2\pi) + i\sin(y+2\pi)] = e^x[\cos y + i\sin y] = e^z$. □

13.5 De Moivre's Formula

THEOREM 13.8. De Moivre's Formula For any integer n and real number θ:

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$$

or equivalently:

$$(e^{i\theta})^n = e^{in\theta}$$

Proof. By induction using the multiplication rule for polar forms. For positive n, repeated application gives $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$. For negative n, use the reciprocal formula. □

13.6 nth Roots of Complex Numbers

THEOREM 13.9. nth Roots For a nonzero complex number $z = r(\cos\theta + i\sin\theta)$ and positive integer n, there are exactly n distinct nth roots given by:

$$z^{1/n} = r^{1/n}\left[\cos\left(\frac{\theta + 2\pi k}{n}\right) + i\sin\left(\frac{\theta + 2\pi k}{n}\right)\right]$$

for $k = 0, 1, 2, \ldots, n-1$.

Proof. We seek $w = \rho(\cos\phi + i\sin\phi)$ such that $w^n = z$. By De Moivre's formula:

$$\rho^n(\cos(n\phi) + i\sin(n\phi)) = r(\cos\theta + i\sin\theta)$$

This requires $\rho^n = r$ (so $\rho = r^{1/n}$) and $n\phi = \theta + 2\pi k$ for integer k. Thus $\phi = (\theta + 2\pi k)/n$. The distinct values occur for $k = 0, 1, \ldots, n-1$. □

14

SEQUENCES AND INFINITE SERIES

14.1 Sequences

DEFINITION 14.1. Sequence A sequence is a function whose domain is the set of positive integers. If f is a sequence, the value $f(n)$ is called the nth term of the sequence and is denoted by a_n or f_n. The sequence is denoted by $\{a_n\}$.

DEFINITION 14.2. Convergence of a Sequence A sequence $\{a_n\}$ is said to converge to a limit L, written $\lim_{n\to\infty} a_n = L$, if for every $\epsilon > 0$ there exists an integer N such that:

$$|a_n - L| < \epsilon \quad \text{whenever } n \geq N$$

If no such L exists, the sequence is said to diverge.

THEOREM 14.1. Uniqueness of Limits If a sequence converges, its limit is unique.

Proof. Suppose $\lim_{n\to\infty} a_n = L_1$ and $\lim_{n\to\infty} a_n = L_2$. For any $\epsilon > 0$, there exists N_1, N_2 such that $|a_n - L_1| < \epsilon/2$ for $n \geq N_1$ and $|a_n - L_2| < \epsilon/2$ for $n \geq N_2$. For $n \geq \max(N_1, N_2)$:

$$|L_1 - L_2| \leq |L_1 - a_n| + |a_n - L_2| < \epsilon/2 + \epsilon/2 = \epsilon$$

Since ϵ is arbitrary, $|L_1 - L_2| = 0$, so $L_1 = L_2$. □

THEOREM 14.2. Properties of Limits of Sequences If $\lim_{n\to\infty} a_n = A$ and $\lim_{n\to\infty} b_n = B$, then:

(a) $\lim_{n\to\infty}(a_n + b_n) = A + B$

(b) $\lim_{n\to\infty}(a_n \cdot b_n) = A \cdot B$

(c) $\lim_{n\to\infty}(ca_n) = cA$ for any constant c

(d) $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{A}{B}$, provided $B \neq 0$

Proof. (a) Given $\epsilon > 0$, choose N such that $|a_n - A| < \epsilon/2$ and $|b_n - B| < \epsilon/2$ for $n \geq N$. Then:

$$|(a_n + b_n) - (A + B)| \leq |a_n - A| + |b_n - B| < \epsilon/2 + \epsilon/2 = \epsilon$$

(b) Since $\{a_n\}$ converges, it is bounded, say $|a_n| \leq M$. Given $\epsilon > 0$, choose N such that for $n \geq N$: $|a_n - A| < \epsilon/(2|B| + 1)$ and $|b_n - B| < \epsilon/(2M)$. Then:

$$\begin{aligned} |a_n b_n - AB| &= |a_n b_n - a_n B + a_n B - AB| \\ &\leq |a_n||b_n - B| + |B||a_n - A| \\ &< M \cdot \frac{\epsilon}{2M} + |B| \cdot \frac{\epsilon}{2|B| + 1} < \epsilon \end{aligned}$$

□

THEOREM 14.3. Sandwich Theorem for Sequences If $a_n \leq b_n \leq c_n$ for all $n \geq N$ for some N, and if $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, then $\lim_{n\to\infty} b_n = L$.

Proof. Given $\epsilon > 0$, there exists N_1 such that $|a_n - L| < \epsilon$ and $|c_n - L| < \epsilon$ for $n \geq N_1$. For $n \geq \max(N, N_1)$:

$$L - \epsilon < a_n \leq b_n \leq c_n < L + \epsilon$$

Thus $|b_n - L| < \epsilon$. □

THEOREM 14.4. Monotone Convergence Theorem A monotone increasing sequence that is bounded above converges. A monotone decreasing sequence that is bounded below converges.

Proof. Let $\{a_n\}$ be monotone increasing and bounded above. Let $L = \sup\{a_n : n \geq 1\}$. For any $\epsilon > 0$, since $L - \epsilon$ is not an upper bound, there exists N such that $a_N > L - \epsilon$. Since $\{a_n\}$ is increasing, for $n \geq N$: $L - \epsilon < a_N \leq a_n \leq L < L + \epsilon$. Thus $|a_n - L| < \epsilon$ for $n \geq N$. □

14.2 Infinite Series

DEFINITION 14.3. Infinite Series Given a sequence $\{a_n\}$, the expression $\sum_{n=1}^{\infty} a_n$ is called an infinite series. The nth partial sum is $s_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^{n} a_k$. The series converges if the sequence of partial sums $\{s_n\}$ converges, and we write $\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} s_n$.

THEOREM 14.5. Necessary Condition for Convergence If $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

Proof. Let $s_n = \sum_{k=1}^{n} a_k$. If the series converges, $\lim_{n\to\infty} s_n = S$ exists. Then $a_n = s_n - s_{n-1}$, so:

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} s_n - \lim_{n\to\infty} s_{n-1} = S - S = 0$$

□

THEOREM 14.6. Linear Property of Series If $\sum_{n=1}^{\infty} a_n = A$ and $\sum_{n=1}^{\infty} b_n = B$, then:

$$\sum_{n=1}^{\infty}(ca_n) = cA \quad \text{and} \quad \sum_{n=1}^{\infty}(a_n + b_n) = A + B$$

Proof. Follows from the corresponding properties for sequences applied to the partial sums. □

THEOREM 14.7. Comparison Test Suppose $0 \leq a_n \leq b_n$ for all $n \geq N$.

(a) If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

(b) If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} b_n$ diverges.

Proof. (a) Let $s_n = \sum_{k=1}^{n} a_k$ and $t_n = \sum_{k=1}^{n} b_k$. For $n \geq N$: $s_n \leq s_N + (t_n - t_N) \leq s_N + T$ where $T = \sum_{k=N+1}^{\infty} b_k$. Thus $\{s_n\}$ is bounded above and increasing, hence convergent.

(b) This is the contrapositive of (a). □

THEOREM 14.8. Limit Comparison Test Let $\sum a_n$ and $\sum b_n$ be series with positive terms. If $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where $0 < c < \infty$, then both series converge or both diverge.

Proof. Since $\lim_{n\to\infty} \frac{a_n}{b_n} = c$, there exists N such that for $n \geq N$:

$$\frac{c}{2} < \frac{a_n}{b_n} < \frac{3c}{2}$$

Thus $\frac{c}{2}b_n < a_n < \frac{3c}{2}b_n$ for $n \geq N$. By the comparison test, $\sum a_n$ converges iff $\sum b_n$ converges. □

THEOREM 14.9. Integral Test Let f be a positive, continuous, decreasing function on $[1, \infty)$ with $f(n) = a_n$. Then $\sum_{n=1}^{\infty} a_n$ converges if and only if $\int_1^{\infty} f(x)\,dx$ converges.

Proof. Since f is decreasing, for $k \geq 2$: $f(k) \leq \int_{k-1}^{k} f(x)\,dx \leq f(k-1)$. Summing from $k = 2$ to n:

$$\sum_{k=2}^{n} f(k) \leq \int_1^n f(x)\,dx \leq \sum_{k=1}^{n-1} f(k)$$

Thus the partial sums $\sum_{k=1}^{n} a_k$ are bounded iff the integrals $\int_1^n f(x)\,dx$ are bounded. By the monotone convergence theorem, both converge or both diverge. □

THEOREM 14.10. Ratio Test Let $\sum a_n$ be a series with positive terms and suppose:

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = L$$

(a) If $L < 1$, the series converges.

(b) If $L > 1$, the series diverges.

(c) If $L = 1$, the test is inconclusive.

Proof. (a) If $L < 1$, choose r with $L < r < 1$. There exists N such that $\frac{a_{n+1}}{a_n} < r$ for $n \geq N$. Thus $a_{N+k} < a_N r^k$ for $k \geq 0$. The geometric series $\sum r^k$ converges, so by comparison, $\sum a_n$ converges.

(b) If $L > 1$, then $a_{n+1} > a_n$ for $n \geq N$, so a_n does not approach 0, and the series diverges. □

THEOREM 14.11. Root Test Let $\sum a_n$ be a series with nonnegative terms and suppose:

$$\lim_{n\to\infty} \sqrt[n]{a_n} = L$$

(a) If $L < 1$, the series converges.

(b) If $L > 1$, the series diverges.

(c) If $L = 1$, the test is inconclusive.

Proof. (a) If $L < 1$, choose r with $L < r < 1$. There exists N such that $\sqrt[n]{a_n} < r$ for $n \geq N$. Thus $a_n < r^n$ for $n \geq N$, and comparison with the geometric series gives convergence.

(b) If $L > 1$, then $a_n > 1$ for infinitely many n, so a_n does not approach 0. □

THEOREM 14.12. Alternating Series Test (Leibniz Criterion) If $\{a_n\}$ is a decreasing sequence of positive numbers with $\lim_{n\to\infty} a_n = 0$, then the alternating series $\sum_{n=1}^{\infty}(-1)^{n+1}a_n$ converges.

Proof. Let s_n be the nth partial sum. Consider the even-indexed partial sums s_{2n}:

$$s_{2n} = (a_1 - a_2) + (a_3 - a_4) + \cdots + (a_{2n-1} - a_{2n})$$

Since $\{a_n\}$ is decreasing, each term in parentheses is nonnegative, so $\{s_{2n}\}$ is increasing. Also:

$$s_{2n} = a_1 - (a_2 - a_3) - \cdots - (a_{2n-2} - a_{2n-1}) - a_{2n} \leq a_1$$

Thus $\{s_{2n}\}$ converges. Similarly, $\{s_{2n+1}\}$ converges to the same limit since $s_{2n+1} = s_{2n} + a_{2n+1} \to L + 0 = L$. □

DEFINITION 14.4. Absolute Convergence A series $\sum a_n$ is said to converge absolutely if $\sum |a_n|$ converges. A series that converges but does not converge absolutely is said to converge conditionally.

THEOREM 14.13. Absolute Convergence Implies Convergence If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

Proof. Define $a_n^+ = \max(a_n, 0)$ and $a_n^- = \max(-a_n, 0)$. Then $0 \leq a_n^+ \leq |a_n|$ and $0 \leq a_n^- \leq |a_n|$. By comparison, both $\sum a_n^+$ and $\sum a_n^-$ converge. Since $a_n = a_n^+ - a_n^-$, we have $\sum a_n = \sum a_n^+ - \sum a_n^-$ converges. □

14.3 Improper Integrals

DEFINITION 14.5. Improper Integral of the First Kind For a function f continuous on $[a, \infty)$, the improper integral is defined as:

$$\int_a^{\infty} f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx$$

If the limit exists and is finite, the integral is said to converge; otherwise, it diverges.

DEFINITION 14.6. Improper Integral of the Second Kind If f is continuous on $(a, b]$ but unbounded near a, define:

$$\int_a^b f(x)\,dx = \lim_{c\to a^+} \int_c^b f(x)\,dx$$

THEOREM 14.14. Comparison Test for Improper Integrals Suppose $0 \leq f(x) \leq g(x)$ for all $x \geq a$.

(a) If $\int_a^{\infty} g(x)\,dx$ converges, then $\int_a^{\infty} f(x)\,dx$ converges.

(b) If $\int_a^{\infty} f(x)\,dx$ diverges, then $\int_a^{\infty} g(x)\,dx$ diverges.

Proof. (a) Let $F(b) = \int_a^b f(x)\,dx$ and $G(b) = \int_a^b g(x)\,dx$. Since $f \leq g$, we have $F(b) \leq G(b)$. If $\int_a^{\infty} g$ converges, $G(b)$ is bounded above, so $F(b)$ is bounded above and increasing, hence converges. □

THEOREM 14.15. p-Test for Improper Integrals The improper integral:

$$\int_1^{\infty} \frac{1}{x^p}\,dx$$

converges if $p > 1$ and diverges if $p \leq 1$.

Proof. For $p \neq 1$:

$$\int_1^b \frac{1}{x^p}\,dx = \left[\frac{x^{1-p}}{1-p}\right]_1^b = \frac{b^{1-p} - 1}{1-p}$$

As $b \to \infty$: if $p > 1$, then $1 - p < 0$, so $b^{1-p} \to 0$, giving convergence to $\frac{1}{p-1}$. If $p < 1$, then $b^{1-p} \to \infty$, giving divergence.

For $p = 1$: $\int_1^b \frac{1}{x}\,dx = \log b \to \infty$ as $b \to \infty$. □

15

SEQUENCES AND SERIES OF FUNCTIONS

15.1 Pointwise Convergence

DEFINITION 15.1. Pointwise Convergence A sequence of functions $\{f_n\}$ defined on a set S is said to converge pointwise to a function f on S if for each $x \in S$:

$$\lim_{n\to\infty} f_n(x) = f(x)$$

That is, for each $x \in S$ and each $\epsilon > 0$, there exists N (depending on x and ϵ) such that $|f_n(x) - f(x)| < \epsilon$ whenever $n \geq N$.

DEFINITION 15.2. Pointwise Convergence of Series A series of functions $\sum_{n=1}^{\infty} f_n(x)$ converges pointwise to $f(x)$ on S if the sequence of partial sums $s_n(x) = \sum_{k=1}^{n} f_k(x)$ converges pointwise to $f(x)$ on S.

15.2 Uniform Convergence

DEFINITION 15.3. Uniform Convergence A sequence of functions $\{f_n\}$ converges uniformly to f on a set S if for every $\epsilon > 0$, there exists N (depending only on ϵ) such that:

$$|f_n(x) - f(x)| < \epsilon \quad \text{for all } n \geq N \text{ and all } x \in S$$

THEOREM 15.1. Uniform Convergence and Continuity If $\{f_n\}$ is a sequence of continuous functions on $[a, b]$ that converges uniformly to f on $[a, b]$, then f is continuous on $[a, b]$.

Proof. Given $\epsilon > 0$, choose N such that $|f_N(x) - f(x)| < \epsilon/3$ for all $x \in [a, b]$. Since f_N is continuous at any $c \in [a, b]$, there exists $\delta > 0$ such that $|f_N(x) - f_N(c)| < \epsilon/3$ whenever $|x - c| < \delta$. Then:

$$\begin{aligned} |f(x) - f(c)| &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(c)| + |f_N(c) - f(c)| \\ &< \epsilon/3 + \epsilon/3 + \epsilon/3 = \epsilon \end{aligned}$$

for $|x - c| < \delta$. Thus f is continuous at c. □

THEOREM 15.2. Uniform Convergence and Integration If $\{f_n\}$ is a sequence of continuous functions on $[a, b]$ that converges uniformly to f on $[a, b]$, then:

$$\lim_{n\to\infty} \int_a^b f_n(x)\, dx = \int_a^b f(x)\, dx$$

Equivalently:

$$\lim_{n\to\infty} \int_a^b f_n(x)\, dx = \int_a^b \lim_{n\to\infty} f_n(x)\, dx$$

Proof. Given $\epsilon > 0$, choose N such that $|f_n(x) - f(x)| < \epsilon/(b-a)$ for all $n \geq N$ and $x \in [a, b]$. Then:

$$\left| \int_a^b f_n(x)\,dx - \int_a^b f(x)\,dx \right| \leq \int_a^b |f_n(x) - f(x)|\,dx < \int_a^b \frac{\epsilon}{b-a}\,dx = \epsilon$$

for $n \geq N$. Thus the limit holds. □

THEOREM 15.3. Weierstrass M-Test Let $\{f_n\}$ be a sequence of functions defined on S, and let $\{M_n\}$ be a sequence of nonnegative constants such that $|f_n(x)| \leq M_n$ for all $x \in S$ and all n. If $\sum_{n=1}^{\infty} M_n$ converges, then $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on S.

Proof. Since $\sum M_n$ converges, by the Cauchy criterion, for $\epsilon > 0$ there exists N such that $\sum_{k=n+1}^{m} M_k < \epsilon$ for $m > n \geq N$. Then for all $x \in S$:

$$\left| \sum_{k=n+1}^{m} f_k(x) \right| \leq \sum_{k=n+1}^{m} |f_k(x)| \leq \sum_{k=n+1}^{m} M_k < \epsilon$$

Thus the series satisfies the uniform Cauchy criterion and converges uniformly. □

15.3 Power Series

DEFINITION 15.4. Power Series A power series is a series of the form:

$$\sum_{n=0}^{\infty} a_n(x-a)^n = a_0 + a_1(x-a) + a_2(x-a)^2 + \cdots$$

where a is the center and $\{a_n\}$ are the coefficients.

THEOREM 15.4. Radius of Convergence For any power series $\sum_{n=0}^{\infty} a_n(x-a)^n$, there exists a number R ($0 \leq R \leq \infty$), called the radius of convergence, such that:

(a) The series converges absolutely for $|x - a| < R$

(b) The series diverges for $|x - a| > R$

(c) The series converges uniformly on any interval $[a - r, a + r]$ where $0 < r < R$

Proof. Let $R = \sup\{|x - a| : \sum |a_n(x-a)^n| \text{ converges}\}$. If $|x - a| < R$, there exists y with $|x - a| < |y - a| \leq R$ where $\sum |a_n(y-a)^n|$ converges. Since $|a_n(x-a)^n| = |a_n(y-a)^n| \cdot |\frac{x-a}{y-a}|^n$ and $|\frac{x-a}{y-a}| < 1$, comparison with the geometric series gives absolute convergence.

If $|x - a| > R$, then $\sum |a_n(x-a)^n|$ diverges by definition of R, so the series diverges.

For uniform convergence on $[a - r, a + r]$ with $r < R$, choose y with $r < |y - a| < R$. Then $\sum |a_n(y-a)^n|$ converges, and for $|x - a| \leq r$: $|a_n(x-a)^n| \leq |a_n|r^n \leq |a_n(y-a)^n|$. By the Weierstrass M-test with $M_n = |a_n(y-a)^n|$, the series converges uniformly. □

THEOREM 15.5. Ratio Test for Radius of Convergence If $\lim_{n\to\infty} |\frac{a_{n+1}}{a_n}| = L$ (or $\limsup_{n\to\infty} |\frac{a_{n+1}}{a_n}| = L$), then the radius of convergence is:

$$R = \begin{cases} \frac{1}{L} & \text{if } 0 < L < \infty \\ \infty & \text{if } L = 0 \\ 0 & \text{if } L = \infty \end{cases}$$

Proof. Apply the ratio test to $\sum |a_n(x-a)^n|$:

$$\lim_{n\to\infty}\left|\frac{a_{n+1}(x-a)^{n+1}}{a_n(x-a)^n}\right| = L|x-a|$$

The series converges when $L|x-a| < 1$, i.e., $|x-a| < 1/L$. □

15.4 Properties of Power Series

THEOREM 15.6. Continuity of Power Series A power series $f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n$ is continuous on its interval of convergence.

Proof. On any $[a-r, a+r]$ with $r < R$, the series converges uniformly by the previous theorem. Since each partial sum is continuous, the limit function f is continuous. Since this holds for any $r < R$, f is continuous on $(a-R, a+R)$. □

THEOREM 15.7. Integration of Power Series If $f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n$ has radius of convergence $R > 0$, then for $|x-a| < R$:

$$\int_a^x f(t)\,dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1}(x-a)^{n+1}$$

The integrated series has the same radius of convergence R.

Proof. On $[a-r, a+r]$ with $r < R$, the series converges uniformly, so:

$$\int_a^x f(t)\,dt = \sum_{n=0}^{\infty}\int_a^x a_n(t-a)^n\,dt = \sum_{n=0}^{\infty}\frac{a_n}{n+1}(x-a)^{n+1}$$

The radius of convergence is the same since $\limsup \sqrt[n]{|a_n|/(n+1)} = \limsup \sqrt[n]{|a_n|}$. □

THEOREM 15.8. Differentiation of Power Series If $f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n$ has radius of convergence $R > 0$, then f is differentiable on $(a-R, a+R)$ and:

$$f'(x) = \sum_{n=1}^{\infty} na_n(x-a)^{n-1}$$

The differentiated series has the same radius of convergence R.

Proof. For $x \in (a-R, a+R)$, choose r with $|x-a| < r < R$. The differentiated series $\sum na_n(x-a)^{n-1}$ converges uniformly on $[a-r, a+r]$ by comparison with $\sum n|a_n|r^{n-1}$ which converges by the ratio test. Thus term-by-term differentiation is valid. The radius is the same since $\limsup \sqrt[n]{n|a_n|} = \limsup \sqrt[n]{|a_n|}$. □

15.5 Taylor Series

DEFINITION 15.5. Taylor Series If f has derivatives of all orders at a, the Taylor series of f at a is:

$$\sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(x-a)^n$$

When $a = 0$, this is called the Maclaurin series.

THEOREM 15.9. Taylor Series Convergence Let f have derivatives of all orders on $(a-R, a+R)$. If there exists M such that $|f^{(n)}(x)| \le M$ for all n and all $x \in (a-R, a+R)$, then the Taylor series converges to $f(x)$ for all $x \in (a-R, a+R)$.

Proof. By Taylor's theorem with remainder:

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + R_n(x)$$

where $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$ for some c between a and x. Since $|f^{(n+1)}(c)| \leq M$:

$$|R_n(x)| \leq \frac{M}{(n+1)!}|x-a|^{n+1} \to 0 \quad \text{as } n \to \infty$$

for $|x-a| < R$. Thus the Taylor series converges to $f(x)$. □

THEOREM 15.10. Uniqueness of Power Series Representation If $f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n$ for $|x-a| < R$, then $a_n = \frac{f^{(n)}(a)}{n!}$ for all n.

Proof. Setting $x = a$ gives $f(a) = a_0$. Differentiating term by term n times:

$$f^{(n)}(x) = \sum_{k=n}^{\infty} k(k-1)\cdots(k-n+1)a_k(x-a)^{k-n}$$

Setting $x = a$, all terms with $k > n$ vanish, leaving $f^{(n)}(a) = n!a_n$. Thus $a_n = f^{(n)}(a)/n!$. □

Part V

Vector Algebra

16

VECTOR ALGEBRA

16.1 The Vector Space of n-tuples of Real Numbers

DEFINITION 16.1. Vector For a positive integer n, an ordered n-tuple of real numbers $(a_1, a_2, \ldots, a_n)$ is called an n-dimensional vector or n-vector. The set of all n-vectors is denoted by V_n.

DEFINITION 16.2. Vector Operations For vectors $A = (a_1, \ldots, a_n)$ and $B = (b_1, \ldots, b_n)$ in V_n and scalar $c \in \mathbb{R}$:

(a) Addition: $A + B = (a_1 + b_1, a_2 + b_2, \ldots, a_n + b_n)$

(b) Scalar multiplication: $cA = (ca_1, ca_2, \ldots, ca_n)$

(c) Zero vector: $O = (0, 0, \ldots, 0)$

(d) Negative: $-A = (-a_1, -a_2, \ldots, -a_n)$

THEOREM 16.1. Properties of Vector Space For all vectors A, B, C in V_n and all scalars c, d:

(a) $A + B = B + A$ (commutativity)

(b) $A + (B + C) = (A + B) + C$ (associativity)

(c) $A + O = A$ (additive identity)

(d) $A + (-A) = O$ (additive inverse)

(e) $c(A + B) = cA + cB$ (distributivity)

(f) $(c + d)A = cA + dA$ (distributivity)

(g) $(cd)A = c(dA)$ (associativity of scalar mult.)

(h) $1A = A$ (scalar identity)

Proof. These follow directly from the corresponding properties of real numbers applied component-wise. □

16.2 The Dot Product

DEFINITION 16.3. Dot Product For vectors $A = (a_1, \ldots, a_n)$ and $B = (b_1, \ldots, b_n)$ in V_n, the dot product is:

$$A \cdot B = \sum_{k=1}^{n} a_k b_k = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$$

THEOREM 16.2. Properties of Dot Product For all vectors A, B, C in V_n and scalar c:

(a) $A \cdot B = B \cdot A$ (symmetry)

(b) $A \cdot (B + C) = A \cdot B + A \cdot C$ (linearity)

(c) $c(A \cdot B) = (cA) \cdot B = A \cdot (cB)$ (homogeneity)

(d) $A \cdot A > 0$ if $A \neq O$, and $A \cdot A = 0$ if $A = O$ (positive definiteness)

Proof. All follow from the definition and properties of real numbers. For (d): $A \cdot A = \sum a_k^2 \geq 0$ with equality iff all $a_k = 0$. □

DEFINITION 16.4. Norm and Length The norm (length) of vector A is $\|A\| = \sqrt{A \cdot A} = \sqrt{\sum_{k=1}^{n} a_k^2}$.

THEOREM 16.3. Cauchy-Schwarz Inequality For all vectors A, B in V_n:

$$|A \cdot B| \leq \|A\| \|B\|$$

with equality if and only if one vector is a scalar multiple of the other.

Proof. If $B = O$, both sides are 0. Otherwise, for any real t:

$$\|A - tB\|^2 = (A - tB) \cdot (A - tB) = \|A\|^2 - 2t(A \cdot B) + t^2 \|B\|^2 \geq 0$$

Setting $t = \frac{A \cdot B}{\|B\|^2}$:

$$\|A\|^2 - \frac{(A \cdot B)^2}{\|B\|^2} \geq 0$$

Thus $(A \cdot B)^2 \leq \|A\|^2 \|B\|^2$, giving $|A \cdot B| \leq \|A\| \|B\|$. Equality holds iff $A = tB$. □

THEOREM 16.4. Triangle Inequality For all vectors A, B in V_n:

$$\|A + B\| \leq \|A\| + \|B\|$$

Proof. Using Cauchy-Schwarz:

$$\begin{aligned} \|A + B\|^2 &= (A + B) \cdot (A + B) = \|A\|^2 + 2(A \cdot B) + \|B\|^2 \\ &\leq \|A\|^2 + 2|A \cdot B| + \|B\|^2 \\ &\leq \|A\|^2 + 2\|A\| \|B\| + \|B\|^2 = (\|A\| + \|B\|)^2 \end{aligned}$$

Taking square roots gives the result. □

16.3 Orthogonality

DEFINITION 16.5. Orthogonal Vectors Two vectors A and B are orthogonal (perpendicular) if $A \cdot B = 0$.

THEOREM 16.5. Pythagorean Theorem Vectors A and B are orthogonal if and only if:

$$\|A + B\|^2 = \|A\|^2 + \|B\|^2$$

Proof. $\|A+B\|^2 = \|A\|^2+2(A\cdot B)+\|B\|^2 = \|A\|^2+\|B\|^2$ if and only if $A\cdot B = 0$. □

16.4 Linear Independence and Bases

DEFINITION 16.6. Linear Combination A vector X is a linear combination of vectors $A_1, A_2, \ldots, A_k$ if:

$$X = c_1A_1 + c_2A_2 + \cdots + c_kA_k$$

for some scalars $c_1, c_2, \ldots, c_k$.

DEFINITION 16.7. Linear Independence Vectors $A_1, A_2, \ldots, A_k$ are linearly independent if:

$$c_1A_1 + c_2A_2 + \cdots + c_kA_k = O \implies c_1 = c_2 = \cdots = c_k = 0$$

Otherwise, they are linearly dependent.

DEFINITION 16.8. Basis A set of vectors $\{E_1, E_2, \ldots, E_n\}$ is a basis for V_n if:

(a) They are linearly independent

(b) Every vector in V_n can be written as a linear combination of $E_1, \ldots, E_n$

The standard basis consists of $E_k = (0, \ldots, 1, \ldots, 0)$ with 1 in the kth position.

THEOREM 16.6. Unique Representation If $\{E_1, \ldots, E_n\}$ is a basis for V_n, then every vector $X \in V_n$ can be written uniquely as:

$$X = x_1E_1 + x_2E_2 + \cdots + x_nE_n$$

Proof. Since $\{E_i\}$ spans V_n, such a representation exists. If $X = \sum x_iE_i = \sum y_iE_i$, then $\sum(x_i - y_i)E_i = O$. By linear independence, $x_i - y_i = 0$ for all i, so the representation is unique. □

16.5 Complex Vector Space $V_n(\mathbb{C})$

DEFINITION 16.9. Complex Vector The complex vector space $V_n(\mathbb{C})$ is the set of all ordered n-tuples of complex numbers $(a_1, a_2, \ldots, a_n)$ where each $a_k \in \mathbb{C}$.

DEFINITION 16.10. Hermitian Dot Product For vectors $A = (a_1, \ldots, a_n)$ and $B = (b_1, \ldots, b_n)$ in $V_n(\mathbb{C})$, the Hermitian dot product is:

$$A \cdot B = \sum_{k=1}^{n} a_k\overline{b_k} = a_1\overline{b_1} + a_2\overline{b_2} + \cdots + a_n\overline{b_n}$$

where $\overline{b_k}$ denotes the complex conjugate of b_k.

THEOREM 16.7. Properties of Hermitian Dot Product For all vectors A, B, C in $V_n(\mathbb{C})$ and complex scalar c:

(a) $A \cdot B = \overline{B \cdot A}$ (conjugate symmetry)

(b) $A \cdot (B + C) = A \cdot B + A \cdot C$ (linearity in first argument)

(c) $c(A \cdot B) = (cA) \cdot B$ and $A \cdot (cB) = \overline{c}(A \cdot B)$ (homogeneity)

(d) $A \cdot A > 0$ if $A \neq O$, and $A \cdot A = 0$ if $A = O$ (positive definiteness)

Proof. (a) $A \cdot B = \sum a_k \overline{b_k} = \sum \overline{\overline{a_k} b_k} = \overline{\sum \overline{a_k} b_k} = \overline{B \cdot A}$.

(c) $(cA) \cdot B = \sum (ca_k)\overline{b_k} = c \sum a_k \overline{b_k} = c(A \cdot B)$. And $A \cdot (cB) = \sum a_k \overline{cb_k} = \sum a_k \overline{c}\overline{b_k} = \overline{c}(A \cdot B)$.

(d) $A \cdot A = \sum a_k \overline{a_k} = \sum |a_k|^2 \geq 0$ with equality iff all $a_k = 0$. □

DEFINITION 16.11. Norm in Complex Space The norm of $A \in V_n(\mathbb{C})$ is $\|A\| = \sqrt{A \cdot A} = \sqrt{\sum_{k=1}^{n} |a_k|^2}$.

THEOREM 16.8. Cauchy-Schwarz Inequality for Complex Vectors For all vectors A, B in $V_n(\mathbb{C})$:

$$|A \cdot B| \leq \|A\| \|B\|$$

Proof. For any complex scalar t, $\|A - tB\|^2 = (A - tB) \cdot (A - tB) \geq 0$. Expanding:

$$\|A\|^2 - t(A \cdot B) - \overline{t}(B \cdot A) + |t|^2 \|B\|^2 \geq 0$$

Setting $t = \frac{A \cdot B}{\|B\|^2}$ (assuming $B \neq O$) and using $A \cdot B = \overline{B \cdot A}$:

$$\|A\|^2 - \frac{|A \cdot B|^2}{\|B\|^2} \geq 0$$

Thus $|A \cdot B|^2 \leq \|A\|^2 \|B\|^2$. □

17

APPLICATIONS TO ANALYTIC GEOMETRY

17.1 Lines

DEFINITION 17.1. Parametric Equation of a Line A line L through point P in the direction of nonzero vector A consists of all points X such that:

$$X = P + tA$$

for some real number t. In coordinates, if $P = (p_1, p_2, p_3)$ and $A = (a_1, a_2, a_3)$, then:

$$x_1 = p_1 + ta_1, \quad x_2 = p_2 + ta_2, \quad x_3 = p_3 + ta_3$$

DEFINITION 17.2. Linear Span of Two Points The line through two distinct points P and Q is given by:

$$L(P, Q) = \{P + t(Q - P) : t \in \mathbb{R}\}$$

THEOREM 17.1. Parallel Lines Two lines $L(P, A)$ and $L(Q, B)$ are parallel if and only if their direction vectors A and B are parallel (i.e., one is a scalar multiple of the other).

Proof. The lines have the same direction if and only if $A = cB$ for some scalar $c \neq 0$. □

17.2 Planes

DEFINITION 17.3. Plane Through a Point with Normal Vector A plane M through point P with nonzero normal vector N consists of all points X satisfying:

$$(X - P) \cdot N = 0$$

or equivalently $X \cdot N = P \cdot N$.

THEOREM 17.2. Plane Through Three Points Given three non-collinear points P, Q, R, there is exactly one plane containing all three. If $A = Q - P$ and $B = R - P$, a normal vector is $N = A \times B$.

Proof. Since P, Q, R are non-collinear, A and B are linearly independent, so $N = A \times B \neq O$. The plane through P with normal N contains Q and R since $(Q - P) \cdot N = A \cdot (A \times B) = 0$ and similarly for R. □

THEOREM 17.3. Distance from Point to Plane The distance from point Q to the plane $(X - P) \cdot N = 0$ is:

$$d = \frac{|(Q - P) \cdot N|}{\|N\|}$$

Proof. The distance is the length of the projection of $(Q - P)$ onto the normal direction:

$$d = \left\| \frac{(Q-P)\cdot N}{\|N\|^2} N \right\| = \frac{|(Q-P)\cdot N|}{\|N\|}$$

□

17.3 The Cross Product

DEFINITION 17.4. Cross Product For vectors $A = (a_1, a_2, a_3)$ and $B = (b_1, b_2, b_3)$ in V_3, the cross product is:

$$A \times B = (a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1)$$

THEOREM 17.4. Properties of Cross Product For all vectors A, B, C in V_3 and scalars c:

(a) $A \times B = -(B \times A)$ (anticommutativity)

(b) $A \times (B + C) = (A \times B) + (A \times C)$ (distributivity)

(c) $(cA) \times B = c(A \times B) = A \times (cB)$ (homogeneity)

(d) $A \times A = O$

(e) $A \cdot (A \times B) = 0$ and $B \cdot (A \times B) = 0$ (orthogonality)

Proof. (a)-(d) follow directly from the definition by component-wise verification.

(e) $A\cdot(A\times B) = a_1(a_2b_3-a_3b_2)+a_2(a_3b_1-a_1b_3)+a_3(a_1b_2-a_2b_1) = 0$ after cancellation. Similarly for B. □

THEOREM 17.5. Geometric Interpretation of Cross Product For vectors A, B in V_3:

$$\|A \times B\| = \|A\|\|B\| \sin\theta$$

where θ is the angle between A and B. The vector $A \times B$ is orthogonal to both A and B, with direction given by the right-hand rule.

Proof. Using $\|A \times B\|^2 + (A \cdot B)^2 = \|A\|^2\|B\|^2$ (Lagrange's identity) and $A \cdot B = \|A\|\|B\| \cos\theta$:

$$\|A \times B\|^2 = \|A\|^2\|B\|^2(1 - \cos^2\theta) = \|A\|^2\|B\|^2 \sin^2\theta$$

Taking square roots gives the result. □

THEOREM 17.6. Scalar Triple Product For vectors A, B, C in V_3:

$$A \cdot (B \times C) = \det \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$$

The absolute value $|A \cdot (B \times C)|$ equals the volume of the parallelepiped determined by A, B, C.

Proof. Expanding both sides gives the same expression. Geometrically, $\|B \times C\|$ is the area of the base parallelogram, and $|A \cdot (B \times C)|/\|B \times C\| = |A||\cos\phi|$ is the height, where ϕ is the angle between A and $B \times C$. □

17.4 Conic Sections

DEFINITION 17.5. Focus-Directrix Definition A conic section is the set of all points P such that the distance from P to a fixed point F (the focus) divided by the distance from P to a fixed line D (the directrix) equals a constant e (the eccentricity):

$$\frac{\|P - F\|}{d(P, D)} = e$$

If $e < 1$: ellipse; if $e = 1$: parabola; if $e > 1$: hyperbola.

THEOREM 17.7. Standard Equation of Ellipse An ellipse with foci at $(c, 0)$ and $(-c, 0)$ and sum of distances $2a$ $(a > c)$ has equation:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

where $b^2 = a^2 - c^2$. The eccentricity is $e = c/a$.

Proof. By definition, $\sqrt{(x-c)^2 + y^2} + \sqrt{(x+c)^2 + y^2} = 2a$. Moving one radical and squaring:

$$(x-c)^2 + y^2 = 4a^2 - 4a\sqrt{(x+c)^2 + y^2} + (x+c)^2 + y^2$$

Simplifying: $-xc = a^2 - a\sqrt{(x+c)^2 + y^2}$. Rearranging and squaring again gives the result. □

THEOREM 17.8. Standard Equation of Parabola A parabola with focus at $(p, 0)$ and directrix $x = -p$ has equation:

$$y^2 = 4px$$

Proof. By definition: $\sqrt{(x-p)^2 + y^2} = |x + p|$. Squaring: $(x-p)^2 + y^2 = (x+p)^2$. Expanding: $x^2 - 2px + p^2 + y^2 = x^2 + 2px + p^2$. Thus $y^2 = 4px$. □

THEOREM 17.9. Standard Equation of Hyperbola A hyperbola with foci at $(c, 0)$ and $(-c, 0)$ and difference of distances $2a$ $(c > a)$ has equation:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

where $b^2 = c^2 - a^2$. The eccentricity is $e = c/a > 1$.

Proof. Similar to the ellipse proof, using $|\sqrt{(x-c)^2 + y^2} - \sqrt{(x+c)^2 + y^2}| = 2a$. □

THEOREM 17.10. Polar Equation of Conic With focus at the origin and directrix $x = d$, a conic with eccentricity e has polar equation:

$$r = \frac{ed}{1 + e\cos\theta}$$

Proof. If P has polar coordinates (r, θ), then $|PF| = r$ and the distance to the directrix $x = d$ is $|d - r\cos\theta|$. By the focus-directrix property: $r = e|d - r\cos\theta|$. For $ed/(1 + e\cos\theta) > 0$, we get $r = ed/(1 + e\cos\theta)$. □

18

CALCULUS OF VECTOR-VALUED FUNCTIONS

18.1 Vector-Valued Functions

DEFINITION 18.1. Vector-Valued Function A vector-valued function $\mathbf{r}$ defined on a set S assigns to each $t \in S$ a unique vector $\mathbf{r}(t)$ in V_n. If $n = 3$, we can write:

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k} = (f(t), g(t), h(t))$$

where f, g, h are real-valued functions called the components of $\mathbf{r}$.
DEFINITION 18.2. Limit of a Vector-Valued Function If $\mathbf{r}(t) = (f_1(t), f_2(t), \ldots, f_n(t))$, then:

$$\lim_{t\to a} \mathbf{r}(t) = \left(\lim_{t\to a} f_1(t), \lim_{t\to a} f_2(t), \ldots, \lim_{t\to a} f_n(t)\right)$$

provided all component limits exist.
DEFINITION 18.3. Continuity A vector-valued function $\mathbf{r}$ is continuous at a if $\lim_{t\to a} \mathbf{r}(t) = \mathbf{r}(a)$.

18.2 Derivatives and Integrals

DEFINITION 18.4. Derivative The derivative of a vector-valued function $\mathbf{r}$ is defined as:

$$\mathbf{r}'(t) = \lim_{h\to 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$

In components, if $\mathbf{r}(t) = (f_1(t), \ldots, f_n(t))$, then $\mathbf{r}'(t) = (f_1'(t), \ldots, f_n'(t))$.
THEOREM 18.1. Differentiation Rules For vector-valued functions $\mathbf{u}, \mathbf{v}$ and scalar function f:

(a) $(\mathbf{u} + \mathbf{v})' = \mathbf{u}' + \mathbf{v}'$

(b) $(f\mathbf{u})' = f'\mathbf{u} + f\mathbf{u}'$

(c) $(\mathbf{u} \cdot \mathbf{v})' = \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}'$

(d) $(\mathbf{u} \times \mathbf{v})' = \mathbf{u}' \times \mathbf{v} + \mathbf{u} \times \mathbf{v}'$ (in V_3)

Proof. All follow from the component-wise definition and the corresponding rules for scalar functions. For (c):

$$(\mathbf{u} \cdot \mathbf{v})' = \sum (u_i v_i)' = \sum (u_i' v_i + u_i v_i') = \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}'$$

□

THEOREM 18.2. Constant Length If $\mathbf{r}(t)$ has constant length ($\|\mathbf{r}(t)\|$ is constant), then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$, i.e., $\mathbf{r}(t)$ is orthogonal to $\mathbf{r}'(t)$.

Proof. If $\|\mathbf{r}(t)\|^2 = \mathbf{r}(t) \cdot \mathbf{r}(t)$ is constant, then:

$$0 = \frac{d}{dt}(\mathbf{r} \cdot \mathbf{r}) = \mathbf{r}' \cdot \mathbf{r} + \mathbf{r} \cdot \mathbf{r}' = 2\mathbf{r} \cdot \mathbf{r}'$$

Thus $\mathbf{r} \cdot \mathbf{r}' = 0$. □

DEFINITION 18.5. Integral The integral of $\mathbf{r}(t) = (f_1(t), \ldots, f_n(t))$ is:

$$\int_a^b \mathbf{r}(t)\, dt = \left(\int_a^b f_1(t)\, dt, \ldots, \int_a^b f_n(t)\, dt \right)$$

18.3 Applications to Curves

DEFINITION 18.6. Tangent Vector For a curve defined by $\mathbf{r}(t)$, the tangent vector at t is $\mathbf{r}'(t)$. The unit tangent vector is:

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$$

THEOREM 18.3. Geometric Interpretation The vector $\mathbf{r}'(t)$ is tangent to the curve at the point $\mathbf{r}(t)$ and points in the direction of increasing t.

Proof. The difference quotient $\frac{\mathbf{r}(t+h)-\mathbf{r}(t)}{h}$ lies along the secant line. As $h \to 0$, this approaches the tangent direction. □

18.4 Arc Length

DEFINITION 18.7. Arc Length The arc length of a curve $\mathbf{r}(t)$ from $t = a$ to $t = b$ is:

$$L = \int_a^b \|\mathbf{r}'(t)\|\, dt$$

THEOREM 18.4. Arc Length is Independent of Parametrization The arc length of a curve depends only on the curve itself, not on the particular parametrization used.

Proof. If $\mathbf{r}_1(u) = \mathbf{r}(t(u))$ where $t(u)$ is a reparametrization, then by the chain rule:

$$\mathbf{r}_1'(u) = \mathbf{r}'(t(u))t'(u)$$

So $\|\mathbf{r}_1'(u)\| = \|\mathbf{r}'(t(u))\||t'(u)|$, and:

$$\int \|\mathbf{r}_1'(u)\|\, du = \int \|\mathbf{r}'(t)\|\, dt$$

by substitution. □

DEFINITION 18.8. Arc Length Function The arc length function $s(t)$ measures length from a fixed point:

$$s(t) = \int_a^t \|\mathbf{r}'(u)\|\, du$$

THEOREM 18.5. $\frac{ds}{dt} = \|\mathbf{r}'(t)\|$.

Proof. By the fundamental theorem of calculus applied to each component. □

18.5 Curvature

DEFINITION 18.9. Curvature The curvature κ of a curve measures how fast the unit tangent vector changes direction:

$$\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\| = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}$$

THEOREM 18.6. Curvature Formula If $\mathbf{r}(t)$ is a smooth curve, then:

$$\kappa = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$$

Proof. Since $\mathbf{T} = \mathbf{r}'/\|\mathbf{r}'\| = \mathbf{r}'/(ds/dt)$, we have $\mathbf{r}' = (ds/dt)\mathbf{T}$. Differentiating:

$$\mathbf{r}'' = \frac{d^2s}{dt^2}\mathbf{T} + \frac{ds}{dt}\mathbf{T}'$$

Since $\mathbf{T}'$ is orthogonal to $\mathbf{T}$ (constant length), taking the cross product:

$$\mathbf{r}' \times \mathbf{r}'' = \left(\frac{ds}{dt}\right)^2 (\mathbf{T} \times \mathbf{T}')$$

Since $\|\mathbf{T}\| = 1$ and $\mathbf{T} \perp \mathbf{T}'$: $\|\mathbf{T} \times \mathbf{T}'\| = \|\mathbf{T}'\|$. Thus:

$$\|\mathbf{r}' \times \mathbf{r}''\| = \left(\frac{ds}{dt}\right)^2 \|\mathbf{T}'\| = \|\mathbf{r}'\|^2\kappa\|\mathbf{r}'\| = \kappa\|\mathbf{r}'\|^3$$

□

18.6 Polar Coordinates

DEFINITION 18.10. Polar Coordinates In polar coordinates, a point is described by (r, θ) where r is the distance from the origin and θ is the angle from the positive x-axis. The relation to Cartesian coordinates is:

$$x = r\cos\theta, \quad y = r\sin\theta$$

THEOREM 18.7. Velocity in Polar Coordinates If $\mathbf{r}(t) = r(t)\mathbf{e}_r$ where $\mathbf{e}_r = (\cos\theta, \sin\theta)$ and $\mathbf{e}_\theta = (-\sin\theta, \cos\theta)$, then:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta$$

where dots denote time derivatives.

Proof. $\mathbf{r} = r\mathbf{e}_r$, so:

$$\dot{\mathbf{r}} = \dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta$$

since $\dot{\mathbf{e}}_r = \dot{\theta}(-\sin\theta, \cos\theta) = \dot{\theta}\mathbf{e}_\theta$. □

THEOREM 18.8. Acceleration in Polar Coordinates The acceleration in polar coordinates is:

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta$$

Proof. Differentiating the velocity formula and using $\dot{\mathbf{e}}_\theta = -\dot{\theta}\mathbf{e}_r$:

$$\begin{aligned}\mathbf{a} &= \ddot{r}\mathbf{e}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}(-\dot{\theta}\mathbf{e}_r) \\ &= (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta\end{aligned}$$

□

18.7 Applications to Planetary Motion

THEOREM 18.9. Kepler's Second Law A particle moving under a central force (force always directed toward a fixed point) sweeps out equal areas in equal times.

Proof. For a central force, the torque is zero, so angular momentum $\mathbf{L} = \mathbf{r} \times m\mathbf{v}$ is constant. In polar coordinates, the area swept out in time dt is $dA = \frac{1}{2}r^2 d\theta$, so:

$$\frac{dA}{dt} = \frac{1}{2}r^2\dot{\theta} = \frac{\|\mathbf{L}\|}{2m} = \text{constant}$$

□

THEOREM 18.10. Kepler's First Law Planets move in elliptical orbits with the sun at one focus.

Proof. Using Newton's law of gravitation $\mathbf{F} = -\frac{GMm}{r^2}\mathbf{e}_r$ and the acceleration formula in polar coordinates, one can derive the orbital equation:

$$r = \frac{L^2/(GMm^2)}{1 + e\cos\theta}$$

which is the polar equation of a conic with eccentricity e. For bound orbits, $e < 1$ giving an ellipse. □

Part VI

Linear Algebra

19

LINEAR SPACES

19.1 Definition of a Linear Space

DEFINITION 19.1. Linear Space (Vector Space) A linear space (or vector space) is a nonempty set V of objects, called elements or vectors, together with two operations called addition and multiplication by scalars (real numbers), subject to the following ten axioms for all vectors $x, y, z \in V$ and all scalars $a, b \in \mathbb{R}$:

(1) Closure under addition: $x + y \in V$

(2) Commutative law: $x + y = y + x$

(3) Associative law: $(x + y) + z = x + (y + z)$

(4) Existence of zero element: There exists $O \in V$ such that $x + O = x$ for all x

(5) Existence of negatives: For each $x \in V$, there exists $-x \in V$ such that $x+(-x) = O$

(6) Closure under scalar multiplication: $ax \in V$

(7) Associative law: $a(bx) = (ab)x$

(8) Distributive law for addition in V: $a(x + y) = ax + ay$

(9) Distributive law for addition of scalars: $(a + b)x = ax + bx$

(10) Identity: $1x = x$

19.2 Examples of Linear Spaces

THEOREM 19.1. Examples The following are examples of linear spaces:

(a) The real vector space V_n of n-tuples of real numbers

(b) The set of all real-valued functions defined on a set S

(c) The set of all continuous functions on an interval $[a, b]$, denoted $C[a, b]$

(d) The set of all polynomials of degree $\leq n$, denoted $\mathcal{P}_n$

(e) The set of all solutions to a homogeneous linear differential equation

19.3 Subspaces

DEFINITION 19.2. Subspace A nonempty subset S of a linear space V is called a subspace if S is itself a linear space under the same operations as V. Equivalently, S is a subspace if:

(a) $O \in S$

(b) S is closed under addition: $x, y \in S \implies x + y \in S$

(c) S is closed under scalar multiplication: $x \in S, a \in \mathbb{R} \implies ax \in S$

19.4 Linear Dependence and Independence

DEFINITION 19.3. Linear Dependence and Independence A finite set $S = \{x_1, x_2, \ldots, x_k\}$ of elements in a linear space V is said to be linearly dependent if there exist scalars $c_1, c_2, \ldots, c_k$, not all zero, such that:

$$\sum_{i=1}^{k} c_i x_i = c_1 x_1 + c_2 x_2 + \cdots + c_k x_k = O$$

If no such relation exists, the set S is called linearly independent.

THEOREM 19.2. A set $S = \{x_1, x_2, \ldots, x_k\}$ is linearly dependent if and only if at least one element of S can be expressed as a linear combination of the others.

Proof. If $\sum c_i x_i = O$ with some $c_j \neq 0$, then $x_j = -\sum_{i \neq j} (c_i/c_j) x_i$. Conversely, if $x_j = \sum_{i \neq j} d_i x_i$, then $\sum_{i \neq j} d_i x_i - x_j = O$ gives a nontrivial relation. □

19.5 Bases and Dimension

DEFINITION 19.4. Span The span of a set S in a linear space V is the set of all finite linear combinations of elements of S:

$$\operatorname{span}(S) = \left\{ \sum_{i=1}^{k} c_i x_i : x_i \in S, c_i \in \mathbb{R}, k \in \mathbb{N} \right\}$$

DEFINITION 19.5. Basis A set B is a basis for a linear space V if:

(a) B is linearly independent

(b) $\operatorname{span}(B) = V$

THEOREM 19.3. Existence of Basis Every finite-dimensional linear space has a basis.

DEFINITION 19.6. Dimension The dimension of a finite-dimensional linear space V, denoted $\dim(V)$, is the number of elements in any basis for V.

THEOREM 19.4. Uniqueness of Representation If $\{e_1, e_2, \ldots, e_n\}$ is a basis for V, then every element $x \in V$ can be written uniquely as:

$$x = \sum_{i=1}^{n} c_i e_i$$

Proof. Since the basis spans V, such a representation exists. If $x = \sum c_i e_i = \sum d_i e_i$, then $\sum (c_i - d_i) e_i = O$. By linear independence, $c_i = d_i$ for all i. □

19.6 Inner Products

DEFINITION 19.7. Inner Product An inner product on a real linear space V is a function $(\cdot,\cdot) : V \times V \to \mathbb{R}$ satisfying:

(a) Symmetry: $(x, y) = (y, x)$ for all $x, y \in V$

(b) Linearity in first argument: $(ax + by, z) = a(x, z) + b(y, z)$ for all $x, y, z \in V$ and $a, b \in \mathbb{R}$

(c) Positive definiteness: $(x, x) \geq 0$ for all $x \in V$, and $(x, x) = 0$ if and only if $x = O$

A linear space with an inner product is called an inner product space.

DEFINITION 19.8. Norm from Inner Product In an inner product space, the norm (length) of a vector x is defined as:

$$\|x\| = \sqrt{(x, x)}$$

THEOREM 19.5. Cauchy-Schwarz Inequality In any inner product space:

$$|(x, y)| \leq \|x\|\|y\|$$

for all $x, y \in V$.

Proof. For any real t, $(x + ty, x + ty) \geq 0$. Expanding:

$$(x, x) + 2t(x, y) + t^2(y, y) \geq 0$$

This quadratic in t is nonnegative for all t, so its discriminant is non-positive:

$$4(x, y)^2 - 4(x, x)(y, y) \leq 0$$

Thus $(x, y)^2 \leq \|x\|^2\|y\|^2$. □

THEOREM 19.6. Triangle Inequality In any inner product space:

$$\|x + y\| \leq \|x\| + \|y\|$$

Proof. $\|x + y\|^2 = (x + y, x + y) = \|x\|^2 + 2(x, y) + \|y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2$. □

19.7 Orthogonality

DEFINITION 19.9. Orthogonality Two elements x and y in an inner product space are orthogonal if $(x, y) = 0$.

THEOREM 19.7. Pythagorean Theorem If x and y are orthogonal, then:

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2$$

Proof. $\|x + y\|^2 = (x + y, x + y) = \|x\|^2 + 2(x, y) + \|y\|^2 = \|x\|^2 + \|y\|^2$. □

THEOREM 19.8. Best Approximation Let S be a finite-dimensional subspace of an inner product space V, and let $\{e_1, \ldots, e_n\}$ be an orthonormal basis for S. For any $x \in V$, the element $s \in S$ that minimizes $\|x - s\|$ is:

$$s = \sum_{i=1}^{n} (x, e_i)e_i$$

This is called the orthogonal projection of x onto S.

Proof. For any $s \in S$, write $s = \sum c_i e_i$. Then:

$$\|x - s\|^2 = (x - \sum c_i e_i, x - \sum c_i e_i) = \|x\|^2 - 2\sum c_i (x, e_i) + \sum c_i^2$$

Completing the square in c_i:

$$\|x - s\|^2 = \|x\|^2 - \sum (x, e_i)^2 + \sum (c_i - (x, e_i))^2$$

This is minimized when $c_i = (x, e_i)$. $\square$

20

LINEAR TRANSFORMATIONS AND MATRICES

20.1 Linear Transformations

DEFINITION 20.1. Linear Transformation Let V and W be linear spaces. A function $T : V \to W$ is called a linear transformation if:

(a) $T(x + y) = T(x) + T(y)$ for all $x, y \in V$ (additivity)

(b) $T(cx) = cT(x)$ for all $x \in V$ and all scalars c (homogeneity)

Equivalently, T is linear if $T(ax + by) = aT(x) + bT(y)$ for all $x, y \in V$ and all scalars a, b.

THEOREM 20.1. Properties of Linear Transformations If $T : V \to W$ is a linear transformation, then:

(a) $T(O_V) = O_W$ (maps zero to zero)

(b) $T(-x) = -T(x)$ for all $x \in V$

(c) $T(\sum_{i=1}^{n} c_i x_i) = \sum_{i=1}^{n} c_i T(x_i)$ for any linear combination

Proof. (a) $T(O_V) = T(0 \cdot O_V) = 0 \cdot T(O_V) = O_W$.
(b) $T(-x) = T((-1)x) = (-1)T(x) = -T(x)$. □

20.2 Null Space and Range

DEFINITION 20.2. Null Space and Range Let $T : V \to W$ be a linear transformation.

(a) The null space (or kernel) of T is $N(T) = \{x \in V : T(x) = O_W\}$

(b) The range (or image) of T is $R(T) = \{T(x) : x \in V\}$

THEOREM 20.2. Null Space and Range are Subspaces Let $T : V \to W$ be a linear transformation. Then:

(a) $N(T)$ is a subspace of V

(b) $R(T)$ is a subspace of W

Proof. (a) If $x, y \in N(T)$, then $T(x+y) = T(x)+T(y) = O+O = O$, so $x+y \in N(T)$. If $x \in N(T)$ and c is a scalar, $T(cx) = cT(x) = cO = O$, so $cx \in N(T)$.
(b) Similar proof for $R(T)$. □

DEFINITION 20.3. Nullity and Rank For a linear transformation $T : V \to W$:

(a) The nullity of T is $\dim N(T)$

(b) The rank of T is $\dim R(T)$

THEOREM 20.3. Rank-Nullity Theorem Let $T : V \to W$ be a linear transformation with V finite-dimensional. Then:

$$\dim V = \text{nullity}(T) + \text{rank}(T)$$

Proof. Let $\{e_1, \ldots, e_k\}$ be a basis for $N(T)$. Extend to a basis $\{e_1, \ldots, e_k, e_{k+1}, \ldots, e_n\}$ for V. We claim $\{T(e_{k+1}), \ldots, T(e_n)\}$ is a basis for $R(T)$.

Spanning: Any $y \in R(T)$ has $y = T(x)$ for some $x = \sum c_i e_i$. Then $y = \sum c_i T(e_i) = \sum_{i=k+1}^n c_i T(e_i)$ since $T(e_i) = O$ for $i \leq k$.

Independence: If $\sum_{i=k+1}^n c_i T(e_i) = O$, then $T(\sum_{i=k+1}^n c_i e_i) = O$, so $\sum_{i=k+1}^n c_i e_i \in N(T)$. Thus $\sum_{i=k+1}^n c_i e_i = \sum_{i=1}^k d_i e_i$, giving $\sum_{i=k+1}^n c_i e_i - \sum_{i=1}^k d_i e_i = O$. By independence of $\{e_i\}$, all $c_i = 0$.

Thus $\dim R(T) = n - k = \dim V - \dim N(T)$. □

20.3 Matrix Representation

DEFINITION 20.4. Matrix of a Linear Transformation Let $T : V \to W$ be a linear transformation with $\dim V = n$ and $\dim W = m$. Let $\{e_1, \ldots, e_n\}$ be a basis for V and $\{f_1, \ldots, f_m\}$ be a basis for W. The matrix of T relative to these bases is the $m \times n$ matrix $A = (a_{ij})$ where:

$$T(e_j) = \sum_{i=1}^m a_{ij} f_i \quad \text{for } j = 1, \ldots, n$$

THEOREM 20.4. Matrix Representation Let $T : V \to W$ have matrix A relative to bases $\{e_j\}$ of V and $\{f_i\}$ of W. If $x = \sum_{j=1}^n x_j e_j \in V$ and $y = T(x) = \sum_{i=1}^m y_i f_i \in W$, then:

$$y_i = \sum_{j=1}^n a_{ij} x_j \quad \text{for } i = 1, \ldots, m$$

In matrix form: $y = Ax$.

Proof. $T(x) = T(\sum_j x_j e_j) = \sum_j x_j T(e_j) = \sum_j x_j \sum_i a_{ij} f_i = \sum_i (\sum_j a_{ij} x_j) f_i$. Comparing with $y = \sum_i y_i f_i$ gives the result. □

20.4 Composition and Matrix Multiplication

DEFINITION 20.5. Composition of Linear Transformations If $T : U \to V$ and $S : V \to W$ are linear transformations, the composition $S \circ T : U \to W$ is defined by:

$$(S \circ T)(x) = S(T(x)) \quad \text{for all } x \in U$$

THEOREM 20.5. Matrix of Composition Let $T : U \to V$ have matrix B and $S : V \to W$ have matrix A (relative to appropriate bases). Then the composition $S \circ T : U \to W$ has matrix AB.

Proof. Let $\dim U = p$, $\dim V = n$, $\dim W = m$. For $x \in U$:

$$(S \circ T)(x) = S(T(x)) = S(Bx) = A(Bx) = (AB)x$$

Thus $S \circ T$ has matrix AB. □

20.5 Invertibility

DEFINITION 20.6. Invertible Linear Transformation A linear transformation $T : V \to W$ is invertible if there exists a linear transformation $T^{-1} : W \to V$ such that:

$$T^{-1} \circ T = I_V \quad \text{and} \quad T \circ T^{-1} = I_W$$

where I_V and I_W are the identity transformations on V and W.

THEOREM 20.6. Invertibility Criteria Let $T : V \to W$ be a linear transformation with V and W finite-dimensional and $\dim V = \dim W = n$. The following are equivalent:

(a) T is invertible

(b) T is injective (one-to-one)

(c) T is surjective (onto)

(d) $\text{rank}(T) = n$

(e) The matrix of T (relative to any bases) is invertible

Proof. By the rank-nullity theorem: $\dim V = \text{nullity}(T) + \text{rank}(T) = n$.
(a) $\Leftrightarrow$ (b): Standard result for any function.
(b) $\Leftrightarrow$ (d): T is injective iff $N(T) = \{O\}$ iff $\text{nullity}(T) = 0$ iff $\text{rank}(T) = n$.
(c) $\Leftrightarrow$ (d): T is surjective iff $R(T) = W$ iff $\text{rank}(T) = \dim W = n$.
(d) $\Leftrightarrow$ (e): The matrix A is invertible iff $Ax = 0$ has only the trivial solution, which is equivalent to $N(T) = \{O\}$. □

20.6 Eigenvalues and Eigenvectors

DEFINITION 20.7. Eigenvalue and Eigenvector Let $T : V \to V$ be a linear transformation. A scalar λ is called an eigenvalue of T if there exists a nonzero vector $x \in V$ such that:

$$T(x) = \lambda x$$

Such a vector x is called an eigenvector of T corresponding to λ.

DEFINITION 20.8. Characteristic Polynomial Let A be the matrix of $T : V \to V$ relative to some basis. The characteristic polynomial of T (and of A) is:

$$p(\lambda) = \det(A - \lambda I)$$

THEOREM 20.7. Eigenvalues and Characteristic Polynomial The eigenvalues of a linear transformation T are the roots of its characteristic polynomial.

Proof. λ is an eigenvalue iff $T(x) = \lambda x$ for some $x \neq O$, iff $(T - \lambda I)(x) = O$ for some $x \neq O$, iff $T - \lambda I$ is not invertible, iff $\det(A - \lambda I) = 0$. □

20.7 Systems of Linear Equations

DEFINITION 20.9. System of Linear Equations A system of m linear equations in n unknowns can be written in matrix form as:

$$Ax = b$$

where A is an $m \times n$ matrix, x is an $n \times 1$ column vector of unknowns, and b is an $m \times 1$ column vector.

THEOREM 20.8. Solution Set Structure Let A be an $m \times n$ matrix. The solution set of the homogeneous system $Ax = O$ is the null space of the linear transformation $T(x) = Ax$, hence is a subspace of $\mathbb{R}^n$.

If x_p is a particular solution of $Ax = b$, then every solution has the form $x = x_p + x_h$ where x_h is a solution of $Ax = O$.

Proof. If $Ax_p = b$ and $Ax = b$, then $A(x - x_p) = O$, so $x - x_p = x_h$ for some x_h with $Ax_h = O$. Thus $x = x_p + x_h$. □

20.8 Gaussian Elimination

THEOREM 20.9. Elementary Row Operations The following elementary row operations on a matrix do not change the solution set of the corresponding linear system:

(a) Interchanging two rows

(b) Multiplying a row by a nonzero scalar

(c) Adding a multiple of one row to another row

THEOREM 20.10. Row Echelon Form By elementary row operations, any matrix can be transformed to row echelon form, from which the solution set of the corresponding linear system can be easily determined.

21

DETERMINANTS

21.1 Introduction

DEFINITION 21.1. Determinant Function Let $A = (a_{ij})$ be an $n \times n$ matrix. The determinant of A, denoted by $\det A$ or $|A|$, is defined inductively as follows:

(a) If $n = 1$, then $\det A = a_{11}$

(b) If $n > 1$, then $\det A = \sum_{j=1}^{n} a_{1j} \cdot \text{cof}(a_{1j})$, where $\text{cof}(a_{1j}) = (-1)^{1+j} \cdot \det A_{1j}$ is the cofactor of a_{1j}, and A_{1j} is the $(n-1) \times (n-1)$ submatrix obtained by deleting the first row and j-th column of A.

21.2 Fundamental Properties of Determinants

THEOREM 21.1. Effect of Row Operations on Determinants Let A be an $n \times n$ matrix.

(a) If B is obtained from A by interchanging two rows, then $\det B = -\det A$

(b) If B is obtained from A by multiplying a row by a scalar c, then $\det B = c \cdot \det A$

(c) If B is obtained from A by adding a multiple of one row to another row, then $\det B = \det A$

Proof. These follow from the definition by induction on n. □

THEOREM 21.2. Zero Row or Column If a matrix A has a row or column of zeros, then $\det A = 0$.

Proof. Expand along the zero row or column. □

THEOREM 21.3. Equal Rows or Columns If a matrix A has two identical rows or two identical columns, then $\det A = 0$.

Proof. If two rows are identical, interchanging them gives the same matrix, so $\det A = -\det A$, hence $\det A = 0$. □

21.3 The Determinant of a Transpose

THEOREM 21.4. Transpose Property For any square matrix A:

$$\det A = \det A^t$$

where A^t is the transpose of A.

Proof. We prove this by induction on n. For $n = 1$, $A = A^t$ so the result is trivial. Assume the theorem holds for matrices of size $(n-1) \times (n-1)$.

Expanding $\det A$ by its first-column minors and $\det A^t$ by its first-row minors, we have:

$$\det A = \sum_{i=1}^{n} (-1)^{i+1} a_{i1} \det A_{i1}, \quad \det A^t = \sum_{j=1}^{n} (-1)^{1+j} a^t_{1j} \det A^t_{1j}$$

But from the definition of transpose we have $a_{i1} = a^t_{1i}$ and $A_{i1} = A^t_{1i}$. Since we are assuming the theorem for determinants of size $(n-1) \times (n-1)$, we have $\det A_{i1} = \det A^t_{1i}$. Hence the foregoing sums are equal term by term, so $\det A = \det A^t$. □

21.4 The Cofactor Matrix

DEFINITION 21.2. Cofactor Matrix Let A be an $n \times n$ matrix. The cofactor matrix of A, denoted cof A, is the matrix whose (i, j) entry is $\text{cof}(a_{ij})$, the cofactor of a_{ij}.

$$(\text{cof } A)_{ij} = (-1)^{i+j} \det A_{ij}$$

where A_{ij} is the submatrix obtained by deleting the i-th row and j-th column of A.

THEOREM 21.5. Cofactor Expansion For any $n \times n$ matrix A with $n \geq 2$:

$$A(\text{cof } A)^t = (\det A)I$$

In particular, if $\det A \neq 0$, the inverse of A exists and is given by:

$$A^{-1} = \frac{1}{\det A}(\text{cof } A)^t$$

Proof. The (i, k) entry of $A(\text{cof } A)^t$ is $\sum_{j=1}^{n} a_{ij} \cdot \text{cof}(a_{kj})$. When $i = k$, this sum equals $\det A$ by the cofactor expansion. When $i \neq k$, this sum equals the determinant of a matrix with two identical rows (the i-th and k-th rows both equal to the i-th row of A), which is 0. Thus $A(\text{cof } A)^t = (\det A)I$. □

21.5 Cramer's Rule

THEOREM 21.6. Cramer's Rule If a system of n linear equations in n unknowns

$$\sum_{j=1}^{n} a_{ij} x_j = b_i \quad (i = 1, 2, \ldots, n)$$

has a nonsingular coefficient matrix $A = (a_{ij})$, then there is a unique solution for the system given by the formulas:

$$x_j = \frac{1}{\det A} \sum_{k=1}^{n} b_k \cdot \text{cof}(a_{kj}) \quad \text{for } j = 1, 2, \ldots, n$$

Equivalently:

$$x_j = \frac{\det C_j}{\det A}$$

where C_j is the matrix obtained from A by replacing the j-th column of A by the column matrix $B = (b_1, \ldots, b_n)^t$.

Proof. The system can be written in matrix form:

$$AX = B$$

where X and B are column matrices. Since A is nonsingular, there is a unique solution given by:

$$X = A^{-1}B = \frac{1}{\det A}(\text{cof } A)^t B$$

The formula follows by equating components.

To derive the equivalent form, note that:

$$\sum_{k=1}^{n} b_k \cdot \text{cof}(a_{kj}) = \det C_j$$

by expanding $\det C_j$ along the j-th column, where C_j is obtained from A by replacing column j with B. $\square$

22

EIGENVALUES AND OPERATORS

22.1 Eigenvalues and Inner Products

DEFINITION 22.1. Hermitian and Skew-Hermitian Operators Let V be a Euclidean space with inner product $(\cdot, \cdot)$. A linear operator $T : V \to V$ is called:

(a) **Hermitian** (or self-adjoint) if $(T(x), y) = (x, T(y))$ for all $x, y \in V$

(b) **Skew-Hermitian** if $(T(x), y) = -(x, T(y))$ for all $x, y \in V$

THEOREM 22.1. Eigenvalues of Hermitian Operators Let $T : V \to V$ be a Hermitian operator on a finite-dimensional Euclidean space V. Then:

(a) All eigenvalues of T are real

(b) Eigenvectors corresponding to distinct eigenvalues are orthogonal

Proof. (a) Let $T(x) = \lambda x$ with $x \neq O$. Then:

$$\lambda(x, x) = (\lambda x, x) = (T(x), x) = (x, T(x)) = (x, \lambda x) = \bar{\lambda}(x, x)$$

Since $(x, x) > 0$, we have $\lambda = \bar{\lambda}$, so λ is real.

(b) Let $T(x) = \lambda x$ and $T(y) = \mu y$ with $\lambda \neq \mu$. Then:

$$\lambda(x, y) = (T(x), y) = (x, T(y)) = (x, \mu y) = \mu(x, y)$$

Since $\lambda \neq \mu$, we have $(x, y) = 0$. □

22.2 Orthogonal and Unitary Transformations

DEFINITION 22.2. Orthogonal Transformation A linear transformation $T : V \to V$ on a real Euclidean space is called orthogonal if it preserves inner products:

$$(T(x), T(y)) = (x, y) \quad \text{for all } x, y \in V$$

Equivalently, T is orthogonal if $\|T(x)\| = \|x\|$ for all $x \in V$.

DEFINITION 22.3. Unitary Transformation A linear transformation $T : V \to V$ on a complex Euclidean space (unitary space) is called unitary if:

$$(T(x), T(y)) = (x, y) \quad \text{for all } x, y \in V$$

THEOREM 22.2. Properties of Orthogonal Transformations Let $T : V \to V$ be an orthogonal transformation on a finite-dimensional real Euclidean space. Then:

(a) T is invertible and T^{-1} is orthogonal

(b) The eigenvalues of T have absolute value 1

(c) If λ is an eigenvalue of T, then $\lambda = \pm 1$

(d) $\det T = \pm 1$

THEOREM 22.3. Orthogonal and Unitary Matrices

(a) A real matrix A is orthogonal if and only if $A^T A = AA^T = I$

(b) A complex matrix A is unitary if and only if $A^* A = AA^* = I$, where A^* is the conjugate transpose

22.3 Spectral Theorem

THEOREM 22.4. Spectral Theorem for Hermitian Operators Let $T : V \to V$ be a Hermitian operator on a finite-dimensional complex Euclidean space. Then there exists an orthonormal basis for V consisting of eigenvectors of T. Equivalently, there exists a unitary matrix U such that:

$$U^* T U = D$$

where D is a real diagonal matrix.

Proof. By induction on $\dim V$. For $\dim V = 1$, the result is trivial. Assume true for spaces of dimension less than n. For $\dim V = n$, let λ_1 be an eigenvalue of T with unit eigenvector u_1. Let $W = \text{span}\{u_1\}$ and $W^\perp$ be the orthogonal complement. Then T maps $W^\perp$ to $W^\perp$, so by induction, $W^\perp$ has an orthonormal basis of eigenvectors. Adding u_1 gives an orthonormal basis for V. □

THEOREM 22.5. Spectral Theorem for Real Symmetric Matrices Let A be a real symmetric $n \times n$ matrix. Then there exists an orthogonal matrix Q such that:

$$Q^T A Q = D$$

where D is a real diagonal matrix containing the eigenvalues of A.

Proof. A real symmetric matrix represents a Hermitian operator on $\mathbb{R}^n$. By the spectral theorem, there exists an orthonormal basis of eigenvectors. The matrix Q with these eigenvectors as columns is orthogonal, and $Q^T A Q = D$. □

22.4 Normal Operators

DEFINITION 22.4. Normal Operator A linear operator $T : V \to V$ on a Euclidean space is called normal if it commutes with its adjoint:

$$TT^* = T^* T$$

where T^* is the adjoint of T defined by $(T(x), y) = (x, T^*(y))$.

THEOREM 22.6. Properties of Normal Operators Let $T : V \to V$ be a normal operator on a finite-dimensional complex Euclidean space. Then:

(a) $\|T(x)\| = \|T^*(x)\|$ for all $x \in V$

(b) T and T^* have the same eigenvectors; if $T(x) = \lambda x$, then $T^*(x) = \bar{\lambda} x$

(c) Eigenvectors corresponding to distinct eigenvalues are orthogonal

THEOREM 22.7. Spectral Theorem for Normal Operators A linear operator T on a finite-dimensional complex Euclidean space is normal if and only if there exists an orthonormal basis for V consisting of eigenvectors of T.

23

LINEAR DIFFERENTIAL EQUATIONS

23.1 Linear Differential Operators

DEFINITION 23.1. Linear Differential Operator Let $C^{(n)}(I)$ denote the linear space of all real-valued functions with n continuous derivatives on an interval I. A linear differential operator L of order n is a linear transformation $L : C^{(n)}(I) \to C(I)$ of the form:

$$L = D^n + a_1 D^{n-1} + \cdots + a_{n-1} D + a_n$$

where $D = \frac{d}{dx}$ is the differentiation operator and $a_1, \ldots, a_n$ are functions in $C(I)$.

THEOREM 23.1. Linearity of L The operator L is a linear transformation.

Proof. Direct verification using the linearity of differentiation. □

23.2 Existence and Uniqueness for Linear Equations

THEOREM 23.2. Existence-Uniqueness Theorem Let $a_1, \ldots, a_n$ and R be continuous on an open interval I, and let $x_0 \in I$. Then for any initial values $b_0, b_1, \ldots, b_{n-1}$, there exists exactly one solution $y = f(x)$ of:

$$L(y) = R(x), \quad f(x_0) = b_0, \quad f'(x_0) = b_1, \quad \ldots, \quad f^{(n-1)}(x_0) = b_{n-1}$$

23.3 The Dimension of the Solution Space

DEFINITION 23.2. Solution Space The set of all solutions of $L(y) = 0$ is called the solution space of L, denoted by $N(L)$.

THEOREM 23.3. Dimension of Solution Space The solution space $N(L)$ of an n-th order linear homogeneous equation has dimension n.

DEFINITION 23.3. Wronskian Let $y_1, \ldots, y_n$ be functions with $n-1$ derivatives on I. The Wronskian is:

$$W(y_1, \ldots, y_n) = \det \begin{pmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix}$$

THEOREM 23.4. Wronskian Criterion Solutions $y_1, \ldots, y_n$ of $L(y) = 0$ are linearly independent iff $W \neq 0$.

23.4 Constant-Coefficient Operators

THEOREM 23.5. Factorization If $p(r) = (r - r_1) \cdots (r - r_n)$ is the characteristic polynomial, then:

$$L = (D - r_1 I) \cdots (D - r_n I)$$

where the factors commute.

THEOREM 23.6. Basis Solutions Let $p(r)$ have roots $r_1, \ldots, r_n$ (with multiplicities). Then:

(a) If r is real with multiplicity m: $e^{rx}, xe^{rx}, \ldots, x^{m-1}e^{rx}$ are solutions.

(b) If $\alpha \pm i\beta$ have multiplicity m: $e^{\alpha x} \cos \beta x, e^{\alpha x} \sin \beta x, \ldots, x^{m-1}e^{\alpha x} \sin \beta x$ are solutions.

23.5 The Nonhomogeneous Equation

THEOREM 23.7. General Solution Let y_p be a particular solution of $L(y) = R(x)$. Then every solution has the form $y = y_p + y_h$ where $L(y_h) = 0$.

23.6 The Method of Undetermined Coefficients

THEOREM 23.8. Annihilator Method If $R(x)$ satisfies $M(R) = 0$ for some operator M, then every solution of $L(y) = R$ satisfies $(M \circ L)(y) = 0$.

24

SYSTEMS OF DIFFERENTIAL EQUATIONS

24.1 Introduction to Systems

DEFINITION 24.1. System of First-Order Equations A system of n first-order differential equations in n unknown functions $y_1, \ldots, y_n$ has the form:

$$y_i' = f_i(x, y_1, \ldots, y_n), \quad i = 1, \ldots, n$$

In vector notation: $\mathbf{y}' = \mathbf{f}(x, \mathbf{y})$ where $\mathbf{y} = (y_1, \ldots, y_n)$.

DEFINITION 24.2. Linear System A system is called linear if it has the form:

$$\mathbf{y}' = A(x)\mathbf{y} + \mathbf{Q}(x)$$

where $A(x)$ is an $n \times n$ matrix and $\mathbf{Q}(x)$ is an n-dimensional vector. The system is homogeneous if $\mathbf{Q}(x) = \mathbf{0}$.

THEOREM 24.1. Existence and Uniqueness for Systems If $A(x)$ and $\mathbf{Q}(x)$ are continuous on an interval I, then for any $x_0 \in I$ and any initial vector $\mathbf{c}$, there exists a unique solution $\mathbf{y}$ of:

$$\mathbf{y}' = A(x)\mathbf{y} + \mathbf{Q}(x), \quad \mathbf{y}(x_0) = \mathbf{c}$$

on the entire interval I.

24.2 Linear Systems with Constant Coefficients

DEFINITION 24.3. Homogeneous Constant-Coefficient System A system of the form $\mathbf{y}' = A\mathbf{y}$ where A is a constant $n \times n$ matrix.

THEOREM 24.2. Solution via Eigenvalues If $\mathbf{v}$ is an eigenvector of A with eigenvalue λ, then $\mathbf{y}(x) = e^{\lambda x}\mathbf{v}$ is a solution of $\mathbf{y}' = A\mathbf{y}$.

Proof. $\mathbf{y}' = \lambda e^{\lambda x}\mathbf{v} = e^{\lambda x}A\mathbf{v} = A(e^{\lambda x}\mathbf{v}) = A\mathbf{y}$. □

THEOREM 24.3. General Solution of Diagonalizable System If A has n linearly independent eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ with eigenvalues $\lambda_1, \ldots, \lambda_n$, then the general solution of $\mathbf{y}' = A\mathbf{y}$ is:

$$\mathbf{y}(x) = c_1 e^{\lambda_1 x}\mathbf{v}_1 + \cdots + c_n e^{\lambda_n x}\mathbf{v}_n$$

24.3 The Matrix Exponential

DEFINITION 24.4. Matrix Exponential For any $n \times n$ matrix A, the matrix exponential e^A is defined by the series:

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots$$

THEOREM 24.4. Convergence of Matrix Exponential The series for e^A converges for all matrices A.

THEOREM 24.5. Properties of Matrix Exponential

(a) $e^0 = I$

(b) $e^A e^B = e^{A+B}$ if $AB = BA$

(c) $(e^A)^{-1} = e^{-A}$

(d) $\frac{d}{dx} e^{xA} = Ae^{xA} = e^{xA}A$

THEOREM 24.6. Solution via Matrix Exponential The unique solution of $\mathbf{y}' = A\mathbf{y}$ with $\mathbf{y}(0) = \mathbf{c}$ is:

$$\mathbf{y}(x) = e^{xA}\mathbf{c}$$

Proof. Differentiating: $\frac{d}{dx} e^{xA}\mathbf{c} = Ae^{xA}\mathbf{c} = A\mathbf{y}$. At $x = 0$: $\mathbf{y}(0) = e^0\mathbf{c} = \mathbf{c}$. □

24.4 Nonhomogeneous Linear Systems

THEOREM 24.7. Variation of Parameters for Systems The general solution of $\mathbf{y}' = A(x)\mathbf{y} + \mathbf{Q}(x)$ is:

$$\mathbf{y}(x) = \Phi(x)\mathbf{c} + \Phi(x) \int \Phi(x)^{-1}\mathbf{Q}(x)\, dx$$

where $\Phi(x)$ is a fundamental matrix (matrix whose columns are linearly independent solutions of the homogeneous system).

Proof. Using the variation of parameters ansatz $\mathbf{y} = \Phi\mathbf{v}$ and substituting yields $\mathbf{v}' = \Phi^{-1}\mathbf{Q}$. □

24.5 Phase Plane Analysis

DEFINITION 24.5. Phase Plane For a 2D system $\mathbf{y}' = A\mathbf{y}$, the phase plane is the (y_1, y_2)-plane where solution curves (trajectories) are plotted.

DEFINITION 24.6. Critical Point A point $\mathbf{y}_0$ where $A\mathbf{y}_0 = \mathbf{0}$ is called a critical point (or equilibrium point).

THEOREM 24.8. Classification of Critical Points For $\mathbf{y}' = A\mathbf{y}$ with $\det A \neq 0$, the origin is a critical point classified by eigenvalues:

(a) Real, distinct, same sign: Node (attracting if negative, repelling if positive)

(b) Real, opposite signs: Saddle point

(c) Complex with nonzero real part: Spiral point

(d) Purely imaginary: Center

(e) Repeated real: Improper node